

Predictive Quantile Regressions under Persistence and Conditional Heteroskedasticity

Rui Fan* Ji Hyung Lee †

August, 2018

Abstract

This paper provides an improved inference for predictive quantile regressions with persistent predictors and conditionally heteroskedastic errors. The confidence intervals based on conventional quantile regression techniques are not valid when predictors are highly persistent. Moreover, the conditional heteroskedasticity introduces rather complicated nuisance parameters in the limit theory, whose estimation errors can be another source of distortion. We propose a size-corrected bootstrap inference thereby avoiding the nuisance parameter estimation. The bootstrap consistency is shown even with the nonstationary predictors and conditionally heteroskedastic innovations. Monte Carlo simulation confirms the significantly better test size performances of the new methods. The empirical exercises on stock return quantile predictability are revisited.

Keywords: α -mixing process, Conditional heteroskedasticity, Moving block bootstrap, Predictive regression, Quantile regression.

JEL classification: C22

1 Introduction

Predictive regression models have been popular in empirical economics. A common example is to infer the predictive relation between financial returns and economic state variables. Nonstationary predictors in predictive regressions typically lead to a spurious inference. The conventional t-test based approaches are not able to correct this type of nonstationary distortion. Extensive studies have been devoted to correct the inflated test size, see e.g., Campbell and Yogo (2006), Kostakis et al. (2014), Phillips and Lee (2013) and more recently, Choi et al. (2016) and Liao et al. (2018).

Quantile regression (QR henceforth; Koenker and Bassett (1978)) is an appealing technique in predictive regressions. We can potentially detect greater predictability at conditional quantiles of

*Department of Economics, Rensselaer Polytechnic Institute, Russell Sage Laboratory 4307, 110 8th Street, Troy, NY 12180. Email: fanr09@gmail.com

†Department of Economics, University of Illinois, 1407 W. Gregory Dr., 214 David Kinley Hall, Urbana, IL 61801. Email: jihyung@illinois.edu

financial returns than at the mean. The standard QR techniques are, however, subject to the same issue of size distortion when predictors are highly persistent. In addition to the predictor persistence, another important issue in predictive regression models is the conditional heteroskedasticity (CHE). We commonly observe the time varying volatility in financial return data so it is important to allow a reasonable form of CHE in the predictive regression errors. The CHE effect, however, has received much less attention in either predictive regression or QR models.

There have been a few studies on QR with persistent regressors. Xiao (2009) developed a novel cointegration framework in nonstationary QR models. In predictive QR models, Maynard et al. (2011, 2018) and Lee (2016) considered Bonferroni-based and IVX-based methods to correct the nonstationary distortion, respectively. The last paper adopts the IVX-filtration idea (Magdalinos and Phillips, 2009) into QR framework and proposed the so-called IVX-QR methods; see Section 2 below for a quick review.

In the nonstationary QR literature, the regression innovation has been typically assumed to be conditionally homoskedastic¹. The validity of standard error estimation in QR inference crucially depends on the assumed regression innovation structure, and the availability of observations near the quantile of interest (Koenker (2005), Ch.3). Under the homoskedastic error assumption, the nuisance parameters in the nonstationary QR limit theory have simpler forms, so we may employ a density estimation using the QR residuals. Oftentimes we may even cancel out the nuisance parameters using some proper normalization. The density estimation is however difficult when observations are sparse in the neighborhood of the quantile of interest. Moreover, when the homoskedastic error assumption does not hold, the corresponding standard error estimation is no longer valid, either by estimation or cancellation.

This paper provides a valid and easy-to-use inference procedure in predictive QR framework with CHE innovations. We use the moving block bootstrap (MBB; Künsch (1989), Liu and Singh (1992) and Fitzenberger (1997) in stationary QR framework) to resample the pairs of the dependent variable and the IVX-filtered predictors. The reduced predictor persistence by IVX filtering enables us to correct the nonstationary distortion in QR. The pairwise MBB then circumvents the issue of nuisance parameter estimation, while preserving the validity via its inherent robustness to heteroskedasticity. The asymptotic validity of this new QR inference is established in the presence of persistent predictor and CHE errors. This *MBB IVX-QR* method therefore provides a convenient size-corrected inference.

Our Monte Carlo study shows that when predictor variables follow various forms of nonstationary processes, the MBB IVX-QR approach not only maintains the good properties of IVX-QR, but also further reduces the inferential errors by avoiding variance estimations, especially at tail quantiles. The results also show that the MBB IVX-QR method provides better finite sample size control under conditional heteroskedasticity. The empirical implication of this improvement is that we can provide a more conservative test for financial return quantile predictability. This is

¹Lee (2016, Remark 2.1) addressed this issue by including a proxy of error variance as a QR predictor; see Section 1.3.3 of the online supplement (Lee, 2014).

confirmed through the empirical exercises.

The paper is organized as follows. Section 2 introduces the model and the econometric issues associated with the QR models with persistent predictors. The section also provides a brief introduction of Lee (2016)'s IVX-QR method. Section 3 studies QR and IVX-QR limit theory under CHE errors, motivating the bootstrap-based inference. Section 4 presents the theory of MBB IVX-QR approach. Section 5 provides a Monte Carlo study to illustrate the improved finite-sample performance of the new methods. Section 6 reinvestigates the empirical exercises on stock return quantile predictability, and Section 7 concludes. All the technical details are relegated to the Appendix, and selected proofs are contained in the Online Appendix.

Notation We use standard notation. $\Rightarrow$ and $\rightarrow_p$ represent convergence in distribution and in probability, respectively. All limit theory assumes $n \rightarrow \infty$ so we sometimes omit this condition. $\sim$ signifies “being distributed as” either exactly or asymptotically depending on the contexts. $O(1)$ and $o(1)$ ($O_p(1)$ and $o_p(1)$) are (stochastically) asymptotically bounded or negligible quantities.

2 Model and Review of Existing Results

We introduce the model and review the econometrics issue and limit theory in nonstationary predictive QR models. Consider the linear predictive QR model:

$$y_t = \beta_{0\tau} + \beta'_{1\tau} x_{t-1} + u_{t\tau} = \beta'_\tau X_{t-1} + u_{t\tau}, \quad (1)$$

where the conditional moment restriction $\Pr(y_t < \beta'_\tau X_{t-1} | \mathcal{F}_{t-1}) = \Pr(u_{t\tau} < 0 | \mathcal{F}_{t-1}) = \tau$ defines the QR parameters $\beta_\tau = (\beta_{0\tau}, \beta'_{1\tau})'$ of $(K+1) \times 1$ vector. Motivated by the stylized fact of CHE of financial asset returns (y_t), we impose CHE to the mean regression innovation $u_t = y_t - \beta'_\mu X_{t-1}$ (equals to y_t under the null hypothesis of no predictability) where $\beta_\mu = (\beta_{0\mu}, \beta'_{1\mu})'$. Then the relation $u_{t\tau} = u_t + (\beta_\mu - \beta_\tau)' X_{t-1}$ clearly indicates the QR innovation $u_{t\tau}$ is also subject to the CHE effects. Without loss of generality, we assume $\beta_\mu = 0$ (no predictability at the mean)² and focus on the significance test $H_0 : \beta_{1\tau} = 0$ for a given $\tau \in (0, 1)$. These tests can detect some quantile specific predictors for financial returns so they are of primary importance in practice. The implied relation between mean and quantile regression innovations is now

$$u_{t\tau} = u_t - \beta_{0\tau} - \beta'_{1\tau} x_{t-1}. \quad (2)$$

Assumption 2.1 *The specification of predictor x_{t-1} follows the autoregressive form*

$$\begin{aligned} x_t &= R_n x_{t-1} + u_{xt}, \\ R_n &= I_K + \frac{C}{n^\alpha}, \text{ for some } \alpha > 0, \end{aligned} \quad (3)$$

²When $\beta_\mu \neq 0$, simply redefining $b_\tau = \beta_\mu - \beta_\tau$ justifies all the following theory with minor modifications.

where n is the sample size and $C = \text{diag}(c_1, c_2, \dots, c_K)$. The predictor innovation u_{xt} satisfies the conditions in Assumption 3.1 below. The pair of (α, C) represents persistences in the multiple predictors of unknown degree, and x_t can belong to any of the following persistence categories³:

I(0) stationary: $\alpha = 0$ and $|1 + c_i| < 1, \forall i$,

MI mildly integrated: $\alpha \in (0, 1)$ and $c_i \in (-\infty, 0), \forall i$,

I(1) local to unity and unit root: $\alpha = 1$ and $c_i \in (-\infty, \infty), \forall i$.

2.1 Review: nonstationary QR distortion

The ordinary QR estimators of parameters are:

$$\hat{\beta}_\tau = \underset{\beta}{\operatorname{argmin}} \sum_{t=1}^n \rho_\tau(y_t - \beta' X_{t-1}), \quad (4)$$

where $\rho_\tau(u) = u(\tau - 1(u < 0))$, $\tau \in (0, 1)$ is the conventional QR loss function.

As shown in Lee (2016), persistence in x_{t-1} can lead to a nonstandard distortion in t-ratio for testing $\beta_{1\tau} = 0$ depending on nonlinear dependence between u_t and u_{xt} , that is:

$$t_{\hat{\beta}_{1\tau}} = \frac{(\hat{\beta}_{1\tau} - \beta_{1\tau})}{s.e(\hat{\beta}_{1\tau})} \sim \underbrace{\left[1 - \lambda(\tau)^2\right]^{1/2} Z}_{\text{standard inference}} + \underbrace{\lambda(\tau) \eta_{LUR}(c)}_{\text{nonstandard distortion}}, \quad (5)$$

where Z and $\eta_{LUR}(c)$ stand for a standard normal distribution and a local to unit root t-statistic, respectively, and $\lambda(\tau) = -\text{cor}(1(u_{t\tau} < 0)u_{xt}) \neq \text{cor}(u_{t\tau}, u_{xt})$.

As the analytical expression (5) shows, the nonstandard distortion becomes greater with (i) smaller $|c|$ and (ii) larger $|\lambda(\tau)|$. Condition (i) is well known from the mean predictive regression literature where the distortion from the highly left-skewed feature of $\eta_{LUR}(c)$ with small $|c|$ has been studied. Condition (ii) is a special feature of QR with persistent regressors.

Therefore, when $\hat{\lambda}(\tau)$ is large in its absolute value (a plug-in estimation is suggested in Lee (2014)) and x_t is highly persistent (smaller c ; suspected from unit root-like behavior), the $t_{\hat{\beta}_{1\tau}}$ can be large even though the true value $\beta_{1\tau}$ is zero, resulting a spurious QR result. Including a spurious predictor can be detrimental in empirical analysis and forecasting.

³Lee (2016) also considered the mildly explosive case. When the regressor is mildly explosive (the signal is extremely strong), the IVX filtration does not alter the stochastic property so the regressor still remains mildly explosive. IVX-QR limit theory for this case comes from the limit theory of mildly explosive process itself, please see Lemma A.1 and A.4 in Lee (2016). MBB IVX-QR Limit theory for this case therefore involves bootstrapping mildly explosive processes. The theory of bootstrap with mildly explosive data is technically demanding (if possible) and seems beyond the scope of this paper.

2.2 Review: IVX filtration and existing limit theory

The IVX method suggested by Magdalinos and Phillips (2009) filters x_t to generate $\tilde{z}_t$ with MI persistence - intermediate between first differencing and the use of levels data. In particular, we choose $F = R_{nz}$ as follows:

$$\tilde{z}_t = R_{nz}\tilde{z}_{t-1} + \triangle x_t, \quad R_{nz} = I_K + \frac{C_z}{n^\delta}, \quad (6)$$

where $\delta \in (0, 1)$, $C_z = c_z I_K$, $c_z < 0$ and $\tilde{z}_0 = 0$.

The parameters $\delta \in (0, 1)$ and $c_z < 0$ are specified by the researcher, and some practical suggestions to use in predictive QR is given in Section 4.1 of Lee (2016).

We define the normalized random variables of $\tilde{z}_{t-1}$ and x_{t-1} by

$$Z_{t-1}^\dagger := k_n^{-1/2}\tilde{z}_{t-1} \text{ and } X_{t-1}^\dagger := k_n^{-1/2}x_{t-1}, \quad (7)$$

where

$$k_n = \begin{cases} I_K & \text{for I(0),} \\ n^{\alpha \wedge \delta} I_K & \text{for MI and I(1).} \end{cases} \quad (8)$$

Following Lee (2016), we also unify the different asymptotic moment matrices for the MI and I(1) cases :

$$V_{cxz} := \begin{cases} V_{zz}^x = \int_0^\infty e^{rC_z} \Omega_{xx} e^{rC_z} dr, & \text{when } \delta \in (0, \alpha \wedge 1), \\ V_{xx} = \int_0^\infty e^{rC} \Omega_{xx} e^{rC} dr, & \text{when } \alpha \in (0, \delta). \end{cases} \quad (9)$$

and

$$\Psi_{cxz} := \begin{cases} -C_z^{-1} \{ \Omega_{xx} + \int dJ_x^c J_x^c \}, & \text{if } \alpha = 1, \\ -C_z^{-1} \{ \Omega_{xx} + C V_{xx} \}, & \text{if } \alpha \in (\delta, 1), \\ V_{cxz} = V_{xx} & \text{if } \alpha \in (0, \delta). \end{cases} \quad (10)$$

where $\Omega_{xx} = E[u_{xt}u'_{xt}]$ and $J_x^c(r) = \int_0^r e^{(r-s)C} dB_x(s)$ is Ornstein-Uhlenbeck (OU) process with $B_x(\cdot)$ being Brownian Motion $BM(\Omega_{xx})$. In order to test $H_0 : \beta_{1\tau} = 0$, Lee (2016) proposed to use QR with $\tilde{z}_{t-1}$ from (6) ⁴:

$$\hat{\beta}_{1\tau}^{IVXQR} = \arg \min_{\beta} \sum_{t=1}^n \rho_{\tau}(y_{t\tau} - \beta_1' \tilde{z}_{t-1}). \quad (11)$$

We have the following asymptotics of null test statistics under homoskedastic mds regression errors.

⁴We use $\hat{\gamma}_{1\tau}^{IVXQR}$ of Theorem 3.2 in Lee (2016) but still write it as $\hat{\beta}_{1\tau}^{IVXQR}$ in this paper for expositional convenience.

Theorem 2.1 (Theorem 3.2 of Lee (2016)) Under $H_0 : \beta_{1\tau} = 0$,

$$(nk_n)^{1/2} \left(\hat{\beta}_{1\tau}^{IVXQR} - \beta_{1\tau} \right) \Rightarrow \begin{cases} N \left(0, \frac{\tau(1-\tau)}{f_{u\tau}(0)^2} \Omega_{xx}^{-1} \right) & \text{for } I(0), \\ N \left(0, \frac{\tau(1-\tau)}{f_{u\tau}(0)^2} V_{cxz}^{-1} \right) & \text{for MI and } I(1), \end{cases}.$$

Remark 2.1 The homoskedasticity of regression errors allows the convenient form of variance-covariance matrix in the limit with a separated sparsity $f_{u\tau}(0)$, which can be either cancelled by a proper normalization or estimated. The stylized facts of financial time series, however, strongly suggest CHE innovations. As we see below, asymptotic theory under heteroskedasticity is quite different, which calls for a new method of statistical inference. The IVX methods with CHE are studied in the mean regression framework (Phillips and Lee (2016) and Kostakis et al (2014)) but the nonstationary QR with CHE has not been addressed in the literature.

3 Limit Theory under Conditional Heteroskedasticity

We study the QR and IVX-QR limit theory with persistent predictors and CHE innovations in this Section. Under CHE errors, the limit theory becomes more involved with complicated forms of nuisance parameters, introducing another potential source of distortion in statistical inference.

3.1 Heuristics

To see the source of the distortion quickly, we assume a univariate $I(1)$ predictor $x_t = (1 + c/n)x_{t-1} + u_{xt}$, with $u_{xt} \sim mds(0, \Sigma_{xx})$. Also assume $u_t = \sigma_t \varepsilon_t$ where ε_t 's are iid mean-zero random sequences, as given in Assumption 3.1 below.

Under $H_0 : \beta_{1\tau} = 0$, from (2), we have $u_{t\tau} = u_t - \beta_{0\tau}$. From (4) with the standard quadratic approximation argument (Pollard (1991) and Knight (1998)),

$$n \left(\hat{\beta}_{1\tau}^{QR} - \beta_{1\tau} \right) \sim \frac{\frac{1}{n} \sum_{t=1}^n \psi_\tau(u_{t\tau}) x_{t-1}}{\frac{1}{n^2} \sum_{t=1}^n f_{u_{t\tau}, t-1}(0) x_{t-1}^2},$$

where $\psi_\tau(u_{t\tau}) = \tau - 1(u_t < \beta_{0\tau})$, and $f_{u_{t\tau}, t-1}(0) = \frac{1}{\sigma_t} f_\varepsilon(F_\varepsilon^{-1}(\tau))$. Note that $\xi_{nt} = \frac{\psi_\tau(u_{t\tau})}{\sqrt{n}}$ is still mds with the quadratic variation of $\tau(1-\tau)$ in spite of CHE $u_{t\tau}$. The standard invariance principle and the convergence results to the stochastic integrals deliver the limit theory of the numerator, $n^{-1} \sum_{t=1}^n \psi_\tau(u_{t\tau}) x_{t-1} \Rightarrow \int \bar{J}_x^c dB_{\psi_\tau}$, where $\bar{J}_x^c = J_x^c(r) - \int_0^1 J_x^c(r) dr$ is the demeaned OU process, and $B_{\psi_\tau} \equiv BM(0, \tau(1-\tau))$. The denominator, however, shows somewhat different asymptotic behavior under a reasonable assumption on σ_t (Assumption 3.1), $n^{-2} \sum_{t=1}^n f_{u_{t\tau}, t-1}(0) x_{t-1}^2 \Rightarrow f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{1}{\sigma_t} \right] \int (\bar{J}_x^c)^2 dr$, leading to

$$n \left(\hat{\beta}_{1\tau}^{QR} - \beta_{1\tau} \right) \sim \frac{1}{f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{1}{\sigma_t} \right]} \frac{\int \bar{J}_x^c dB_{\psi_\tau}}{\int (\bar{J}_x^c)^2 dr}.$$

Following Lee (2016), use the decomposition $dB_{\psi_\tau} = dB_{\psi_\tau \cdot x} + \Sigma_{\psi_\tau x} \Sigma_{xx}^{-1} dB_x$ to have

$$n \left(\hat{\beta}_{1\tau}^{QR} - \beta_{1\tau} \right) \sim \frac{1}{f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{1}{\sigma_t} \right]} \left[\frac{\int \bar{J}_x^c dB_{\psi_\tau \cdot x}}{\int (\bar{J}_x^c)^2} + \left(\frac{\Sigma_{\psi_\tau x}}{\Sigma_{xx}} \right) \frac{\int \bar{J}_x^c dB_x}{\int (\bar{J}_x^c)^2} \right].$$

In addition to the nonstationary distortion arising from $\left(\int (\bar{J}_x^c)^2 dr \right)^{-1} \int \bar{J}_x^c dB_x$ when $\Sigma_{\psi_\tau x}$ is not negligible, we also have an additional nuisance parameter $f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{1}{\sigma_t} \right]$ which is harder to estimate than in the case with conditional homoskedastic errors. Estimation of this nuisance parameter is not easy unless we have both (i) a simple form of DGP of σ_t and (ii) plenty of observations around the quantile of interest. As we see below, the complication of nuisance parameters in the limit theory becomes even worse for MI-I(0) cases, clearly motivating the usage of a time series bootstrap procedure combined with the IVX filtration.

3.2 Limit Theory

To lay out the nonstationary QR limit theory under CHE, let us impose a reasonable weak dependence and moment conditions for the error processes and their conditional variances. Following standard notation, let $\mathcal{F}_{[-\infty, m]} = \sigma(\dots, x_{m-1}, x_m)$ and $\mathcal{F}_{[m+j, \infty]} = \sigma(x_{m+j}, x_{m+j+1}, \dots)$, $j > 0$, denoting for the σ -algebras generated by $\{x_t\}$. The mixing coefficients $\alpha(j)$ are defined as:

$$\alpha(j) = \sup_m \sup_{A \in \mathcal{F}_{[-\infty, m]}, B \in \mathcal{F}_{[m+j, \infty]}} |P(A)P(B) - P(AB)|.$$

Recall when $\alpha(j) \rightarrow 0$ as $j \rightarrow \infty$, we denote $\{x_t\}_{t=1, \dots, n}$ as an α -mixing process. Let $\sigma_t^2 = E(u_t^2 | \mathcal{F}_{t-1})$, $\sigma_{xt}^2 = E(u_{xt}^2 | \mathcal{F}_{t-1})$, $\mathbf{u}_t = (u_t, u_{xt})'$ and $\boldsymbol{\sigma}_t^2 = (\sigma_t^2, \sigma_{xt}^2)$.

Assumption 3.1 1. $\boldsymbol{\sigma}_t^2 \in \mathcal{F}_{t-1}$ and $E(\mathbf{u}_t | \mathcal{F}_{t-1}) = 0$. In particular, $u_t = \sigma_t \varepsilon_t$ where ε_t 's are iid mean-zero random sequences with CDF F_ε and pdf f_ε .

2. The processes $\{\mathbf{u}_t, \boldsymbol{\sigma}_t\}$ are α -mixing sequences of size $\frac{-r}{r-2}$ for $r > 2$, i.e., the mixing coefficients $\alpha(j) = O(j^\lambda)$ for $\lambda < \frac{-r}{r-2} < 0$.

3. $E|u_{xt}|^{2r+\gamma} < C_1$ for all $t = 1, \dots, n$ and some small $\gamma > 0$.

4. $E|\sigma_{xt}^2|^r < \infty$ for all $t = 1, \dots, n$.

Remark 3.1 As in Carrasco and Chen (2002) or Lindner (2009), various GARCH and Stochastic Volatility models have β -mixing (hence α -mixing) properties with exponential decay rates. Indeed, α -mixing processes are bigger classes than just CHE processes, and thus are general enough to include many practical nonlinear time series innovations. To accommodate CHE in the mean predictive regressions, Phillips and Lee (2016) allowed a rather different nonlinear process based on Wu (2005), whereas Kostakis et al. (2014) used of vec-GARCH(p, q) models. The mixing condition imposes a reasonable weak dependence structure for the conditional volatility processes

σ_t but allows their DGP to remain unspecified. A related paper, Portnoy (1991), allowed another form of heteroskedastic and potentially nonstationary (m -decomposable) error process in QR, and derived a Bahadur representation.

Theorem 3.1 (QR limit theory under CHE) Define the normalizing matrices

$$D_n = \begin{cases} \sqrt{n}I_{K+1} & \text{for } I(0), \\ \text{diag}(\sqrt{n}, n^{\frac{1+\alpha}{2}}) & \text{for } MI, \\ \text{diag}(\sqrt{n}, nI_K) & \text{for } I(1). \end{cases}$$

Under Assumptions 2.1 and 3.1, and $H_0 : \beta_{1\tau} = 0$,

$$D_n \left(\hat{\beta}_{\tau}^{QR} - \beta_{\tau} \right) \Rightarrow \begin{cases} N \left(0, \frac{\tau(1-\tau)}{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))^2} \Omega_{\beta_{\tau}} \right) & \text{for } I(0), \\ N \left(0, \frac{\tau(1-\tau)}{\{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))E\left[\frac{1}{\sigma_t}\right]\}^2} \begin{bmatrix} 1 & 0 \\ 0 & V_{xx}^{-1} \end{bmatrix} \right) & \text{for } MI, \\ \frac{1}{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))E\left[\frac{1}{\sigma_t}\right]} \begin{bmatrix} 1 & \int J_x^c(r)' \\ \int J_x^c(r) & \int J_x^c(r)J_x^c(r)' \end{bmatrix}^{-1} \begin{bmatrix} B_{\psi_{\tau}}(1) \\ \int J_x^c(r)dB_{\psi_{\tau}} \end{bmatrix} & \text{for } I(1), \end{cases}$$

where $\Omega_{\beta_{\tau}} = E \left[\frac{X_{t-1}X'_{t-1}}{\sigma_t} \right]^{-1} E \left[X_{t-1}X'_{t-1} \right] E \left[\frac{X_{t-1}X'_{t-1}}{\sigma_t} \right]^{-1}$ and $V_{xx} = \int_0^{\infty} e^{rC} \Omega_{xx} e^{rC} dr$.

Remark 3.2 The signal strength of the CHE effect from σ_t and that of $I(0)$ regressor x_t are comparable, so the nuisance parameter $\Omega_{\beta_{\tau}}$ shows some interaction between them. As the regressor persistence increases to MI - $I(1)$, the signal strength of σ_t is dominated by the regressor persistence so the average CHE effect $E \left[\frac{1}{\sigma_t} \right]$ is separated out in the limit. In all cases, the direct estimation of the standard error is not feasible unless we assume the exact (and convenient) form of CHE (see Koenker and Zhao (1996) and Xiao and Koenker (2009) for ARCH/GARCH specifications). Even with a convenient CHE specification, the tail inference may not be reliable due to the difficulty of density estimation. All these aspects, again, motivate the bootstrap inference in the next Section.

Remark 3.3 From the above argument we confirm that the CHE effect can potentially lead to a severe size distortion, in addition to the nonstationary QR distortion. Roughly speaking, when there is a substantial amount of heteroskedasticity, $E \left[\frac{1}{\sigma_t} \right]$ will shrink thereby inflating the dispersion of the asymptotic distribution in Theorem (3.1).

To remove the nonstationary distortion, we can employ IVX-QR from (11):

Theorem 3.2 (IVX-QR limit theory under CHE) Under $H_0 : \beta_{1\tau} = 0$,

$$(nk_n)^{1/2} \left(\hat{\beta}_{1\tau}^{IVXQR} - \beta_{1\tau} \right) \Rightarrow \begin{cases} N \left(0, \frac{\tau(1-\tau)}{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))^2} \Omega_{\beta_{\tau}} \right) & \text{for } I(0), \\ N \left(0, \frac{\tau(1-\tau)}{\{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))E\left[\frac{1}{\sigma_t}\right]\}^2} V_{cxz}^{-1} \right) & \text{for } MI \text{ and } I(1), \end{cases}.$$

where $\Omega_{\beta_\tau} = E \left[\frac{X_{t-1}X'_{t-1}}{\sigma_t} \right]^{-1} E [X_{t-1}X'_{t-1}] E \left[\frac{X_{t-1}X'_{t-1}}{\sigma_t} \right]^{-1}$ and

$$V_{cxz} := \begin{cases} V_{zz}^x = \int_0^\infty e^{rC_z} \Omega_{xx} e^{rC_z} dr, & \text{when } \delta \in (0, \alpha \wedge 1), \\ V_{xx} = \int_0^\infty e^{rC} \Omega_{xx} e^{rC} dr, & \text{when } \alpha \in (0, \delta). \end{cases}.$$

Remark 3.4 *IVX-filtration is used to achieve asymptotic normality for all $I(0)$ - $I(1)$ parameter spaces. The asymptotic standard error of this IVX-QR estimator under CHE innovation is still hard to estimate. Using the estimated $\hat{f}_{u_\tau}(0)$ under iid limit theory severely distorts the inference⁵, see Table 1 below. Therefore, we study a moving block bootstrap (MBB) based approach instead.*

4 Bootstrap-based Inference: MBB IVX-QR

As we explained above, estimating the nuisance parameters in Theorem (3.2) directly is not easy under practical scenarios. We study the moving block bootstrap (MBB) inference⁶ in this Section, and prove the validity of the proposed test.

Let b be an integer block length and let $\mathcal{B}(t) = (w_t, w_{t+1}, \dots, w_{t+b-1})$ denote a data block with starting point $t \in \{1, \dots, n - b + 1\}$, where $w_t = (y_t, \tilde{z}'_t)'$ and $\tilde{z}_t$ denotes the IVX-filtered regressors from (6). The total number of possible blocks and the number of blocks in one bootstrapped sample are denoted by q and m , respectively. The letters n and ℓ indicate the sample size and the bootstrapped sample size, respectively. Therefore, $n = q + b - 1$ and $\ell = mb$. The MBB procedure is sampling m number of blocks *randomly* with replacement from $\{\mathcal{B}(t) : t = 1, \dots, n - b + 1\}$ yielding MBB sample. This MBB sample $w_1^*, \dots, w_\ell^*$ is defined as $(\mathcal{B}(I_1), \dots, \mathcal{B}(I_m))$ where I_i 's are *iid* discrete uniform variables on $\{1, \dots, n - b + 1\}$. Let P^* , E^* and Var^* denote probability, expectation and variance of the bootstrap distribution conditional on the original sample.

Based on MBB sample $(w_1^*, \dots, w_\ell^*)$, we estimate

$$\hat{\beta}_\tau^{IVXQR*} = \arg \min_{\beta} \sum_{t=1}^{\ell} \rho_\tau (y_t^* - \beta' \tilde{z}_{t-1}^*), \quad (12)$$

which we denote as *MBB IVX-QR* estimator. We prove the first-order asymptotic validity of the MBB IVX-QR under the following rate conditions:

Assumption 4.1 (a) $\ell = O(n)$ for $n \rightarrow \infty$, (b) $k_n = o(n)$, and (c) $b = O(k_n)$ and $b \rightarrow \infty$.

⁵Under the *iid* error assumption, Koenker and Bassett (1982) estimated the sparsity parameters using the histogram methods of Boneva, Kendall, and Stefanov (1971). Welsh (1987) developed a kernel approach estimating the same quantity as a weighted average of Siddiqui estimates (Siddiqui (1960)). In a more general setting of non-iid errors, Hendricks and Koenker (1992) suggested estimating the asymptotic covariance matrix by an extension of the sparsity estimation methods in *iid* setting. Powell (1991) proposed an estimation method following the idea of kernel density estimation. Although these methods help estimating standard errors for statistical inference, the performance of produced statistics are affected by the precision of estimated nuisance parameters and the imposed error structures.

⁶The moving block bootstrap (MBB) are independently formulated by Künsch (1989) and Liu and Singh (1992). Existing literature provides several variants such as the non-overlapping block bootstrap (NBB) by Carlstein (1986), the circular block bootstrap (CBB) and the stationary bootstrap (SB) of Politis and Romano (1992, 1994). We mainly focus on the overlapping block bootstrap method, which we continue to denote as MBB.

Remark 4.1 The rate condition in (a) is standard in MBB literature. In (b), $k_n = o(n)$ is valid by IVX construction (8). (c) $b = O(k_n)$ is a technical condition to derive Lemma A.4 below which is essential to prove Theorem 4.2. This last condition is a sufficient condition, used to bound the various cross-products of higher moments of IVX variables (see Proof of Lemma A.4). Intuitively, the degree of dependence in MBB sample (block length: b) needs to be balanced with the persistence of IVX-filtered data (k_n).

Recall that $Z_{t-1}^\dagger = k_n^{-1/2} \tilde{z}_{t-1}$ and $X_{t-1}^\dagger = k_n^{-1/2} x_{t-1}$. Define $m_t^\dagger = \frac{\tilde{z}_{t-1}}{\sqrt{k_n}} \psi_\tau(u_{t\tau}) = Z_{t-1}^\dagger \psi_\tau(u_{t\tau})$, and $m_t^{*\dagger}$ is similarly defined using MBB sample. We have the following result on the sample mean of $m_t^\dagger$ and $m_t^{*\dagger}$.

Theorem 4.1 (First-order asymptotic validity of the bootstrap score functions). Under Assumptions 2.1, 3.1 and 4.1,

$$\sup_{x \in R} \left| P^*(\sqrt{\ell}(\bar{m}_\ell^{*\dagger} - \bar{m}_n^\dagger) \leq x) - P(\sqrt{n}(\bar{m}_n^\dagger - E m_t^\dagger) \leq x) \right| \rightarrow_p 0, \quad \text{as } n \rightarrow \infty, \quad (13)$$

where $\bar{m}_n^\dagger = \frac{1}{n} \sum_{t=1}^n m_t^\dagger$ and $\bar{m}_\ell^{*\dagger} = \frac{1}{\ell} \sum_{t=1}^\ell m_t^{*\dagger}$.

Remark 4.2 To the best of our knowledge, there exist no bootstrap consistency results for this type of nonlinear statistics of mildly integrated processes. Kim and Nordman (2011) studied the mean of long memory processes, but the relation between mild integration and long memory property has not yet been clearly understood. See Lee (2014, Section 1.7) for a related discussion.

We now state the main theorem of this paper, enabling the MBB percentile methods as discussed in, for example, by Efron and Tibshirani (1993, Ch 13).

Theorem 4.2 (MBB IVX-QR consistency) Under $H_0 : \beta_{1\tau} = 0$, $(\ell k_n)^{1/2}(\hat{\beta}_{1\tau}^{IVXQR*} - \hat{\beta}_{1\tau}^{IVXQR})$ under P^* and $(n k_n)^{1/2}(\hat{\beta}_{1\tau}^{IVXQR} - \beta_{1\tau})$ under P have the same limit distribution H_{β_τ} , where

$$H_{\beta_\tau} \equiv \begin{cases} N\left(0, \frac{\tau(1-\tau)}{f_\varepsilon(F_\varepsilon^{-1}(\tau))^2} \Omega_{\beta_\tau}\right) & \text{for } I(0), \\ N\left(0, \frac{\tau(1-\tau)}{\{f_\varepsilon(F_\varepsilon^{-1}(\tau))E[\frac{1}{\sigma_t^2}]\}^2} V_{cxz}^{-1}\right) & \text{for MI and } I(1), \end{cases} \quad (14)$$

whose nuisance parameters are defined in Theorem 3.2.

Remark 4.3 One common intuition in the bootstrap literature is that if we have asymptotic normality of the original test statistics, the bootstrap consistency will be achieved since the (blockwise) iid resampling easily attains CLT. The reduced persistence of IVX-filtered regressors ($\tilde{z}_t$) is therefore essential to achieve MBB IVX-QR validity. Without this filtration MBB will not be valid, which is illustrated in the Simulation Section below.

5 Monte Carlo Simulation

This section compares the finite sample performances of (a) conventional t-tests (using estimated density), (b) MBB (without IVX correction), (c) IVX-QR of Lee (2016), (d) IVX-QR-HAC, and (e) MBB IVX-QR. The IVX-QR-HAC is a IVX-QR method using the HAC type robust standard error estimator. That is, $\text{var}(1/\sqrt{n} \sum_{t=1}^n x_t \psi_\tau(u_t))$ is estimated using the τ -weighted signs of the quantile regression residuals (see Fitzenberger (1997, Section 4)):

$$\tau 1(\hat{u}_t > 0) - (1 - \tau) 1(\hat{u}_t < 0).$$

For convenience, our setup and notation follow Lee (2016).

The samples are generated from a predictive QR model:

$$y_t = \beta_{0\tau} + \beta_{1\tau} x_{t-1} + u_{t\tau}, \quad (15)$$

$$x_t = \mu_x + r_n x_{t-1} + u_{xt}, \quad (16)$$

where $\beta_{0\tau}$, $\beta_{1\tau}$, μ_x , and r_n are scalars. Let $\beta_{0\tau} = F_{u_t}^{-1}(\tau)$, $\beta_{1\tau} = 0$, and $\mu_x = 0$. Since parameter $r_n = 1 + c/n$ controls the degree of persistence in predictor x_t , we allow the value of $c \in \mathcal{C}$ to vary from -70 to 0 , and $\mathcal{C} = (0, -2, -5, -7, -70)$, where $c = 0$ indicates the exact unit-root process. The larger the value of $|c|$ is, the less persistent the variable x_t is.

We consider two scenarios for the regression errors, $u_{t\tau} \equiv u_t - F_{u_t}^{-1}(\tau)$:

1. $u_t \sim iid$ Gaussian:

$$\begin{pmatrix} u_t & u_{xt} \end{pmatrix}' \sim iid F_u \left(\mathbf{0}_{2 \times 1}, \Sigma_{2 \times 2} \right),$$

where F_u is a bivariate normal distribution. The correlation matrix Σ is standardized as

$$\Sigma = \begin{pmatrix} 1 & \phi \\ \phi & 1 \end{pmatrix}, \quad \phi = -0.95.$$

2. $u_t \sim ARCH(1)$:

$$u_t = \begin{cases} 0 & \text{for } t = 0, \\ \sigma_t \epsilon_t & \text{for } t \geq 1, \end{cases}$$

where $\sigma_t^2 = \gamma_0 + \gamma_1 u_{t-1}^2$. Let $\gamma_0 = 1$ and $\gamma_1 = 0.9$. The errors ϵ_t and u_{xt} are jointly generated by

$$\begin{pmatrix} \epsilon_t \\ u_{xt} \end{pmatrix} \sim iid F_u \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \phi' \\ \phi' & 1 \end{pmatrix} \right), \quad \phi' = -0.9.$$

To apply the IVX-QR technique, we obtain instruments from (6): $\tilde{z}_t = R_{nz} \tilde{z}_{t-1} + \Delta x_t$, $R_{nz} = I_K + n^{-\delta} C_z$, where $\delta \in (0, 1)$, $C_z = c_z I_K$, $c_z < 0$ and $\tilde{z}_0 = 0$. We choose values of (C_z, δ) by the practical rule in section 4.1 of Lee (2016). The essential idea is to choose $c(\delta, n) = n^{1-\delta} c_z$ such that the nonstandard local unit root t-statistics $\eta_{LUR}(c(\delta, n))$ (Phillips, 1987) behaves like $Z \sim \mathcal{N}(0, 1)$

for a given n . Inference on $\hat{\beta}_{1\tau}$ can therefore be based on critical values from the standard normal distribution.

We test $H_0 : \beta_{1\tau} = 0$ versus the local alternatives $H_1 : \beta_{1\tau} = b/n$. Each test is investigated at eleven quantiles from $\tau = 0.05$ to 0.95 with a nominal size of 5% . In each experiment, 1000 samples are randomly generated for $n = 200$ and 700 .

The inferential methods could be classified into two categories: (1) Tests without IVX correction; these include t test and percentile method using MBB technique (Efron and Tibshirani (1993), Koenker (2005)). These types of tests are commonly used in the literature of QR. If x_t is not persistent, the regular QR asymptotics works for t test. Also, the percentile bootstrap method is often employed when the covariance matrix cannot be computed easily. For the block length of MBB, we adopt the form of $constant \times n^{1/4}$ suggested in Hall, Horowitz and Jing (1995) and Lahiri (1999, 2005)⁷. (2) Tests with IVX correction; these types of tests includes IVX-QR, IVX-QR-HAC, and MBB IVX-QR. These tests aim to correct the size distortion caused by persistence in x_t and nonlinear dependence of $u_{t\tau}$ and u_{xt} . We expect that MBB IVX-QR can have better size control than IVX-QR and IVX-QR-HAC, especially when (i) we are interested in inference at tails, and (ii) the covariance matrix is hard to compute, for example, with conditional heteroskedastic errors. Note that although IVX-QR-HAC uses a HAC type robust estimator, the issue of sparsity function estimation, discussed in Theorem 3.2 and Remark 3.4, remains unresolved. Thus we expect better performance of MBB IVX-QR, which is confirmed through the simulation results below.

The MBB IVX-QR test here is conducted as follows:

Procedure of MBB IVX-QR

1. Given data (y_t, x_t) , for $t = 1, \dots, n$, we choose values of (C_z, δ) by the practical rule of Lee (2016). Then construct instruments $\tilde{z}_t$ following (6).
2. Set block length $b = \lceil n^{1/4} \rceil$,⁸ where $\lceil a \rceil$ denotes the least integer that is greater or equal to a . Let $m = \lceil n/b \rceil$. Randomly sample m data blocks from $(\mathcal{B}(1), \dots, \mathcal{B}(q))$, where $\mathcal{B}(t) = (w_t, w_{t+1}, \dots, w_{t+b-1})$ with $w(t) = (y_t, \tilde{z}_t)$, $q = n - b + 1$. Denote the sampled data blocks by $(\mathcal{B}^*(1), \dots, \mathcal{B}^*(m))$ and obtain a resampled data $(w_1^*, \dots, w_\ell^*)$, $\ell = mb$ and $w_t^* = (y_t^*, \tilde{z}_t^*)$.
3. Estimate

$$\hat{\beta}_\tau^* = \arg \min_{\beta} \sum_{t=1}^{\ell} \rho_\tau(y_t^* - \beta' \tilde{z}_{t-1}^*). \quad (17)$$

4. Repeat step 2 and 3 for N times and obtain $(\hat{\beta}_\tau^{*(1)}, \dots, \hat{\beta}_\tau^{*(N)})$.

⁷Hall et al. (1995) studied optimal block size of bootstrap for strictly stationary dependent data, suggesting $n^{1/3}$, $n^{1/4}$, and $n^{1/5}$ under the different contexts. Lahiri (1999, 2005) suggested a nonparametric plug-in method. There is no known result for the block length choice for bootstrapping mildly integrated processes to the best of our knowledge. Thus we follow the standard suggestion.

⁸Under Assumption 4.1, a rate condition $b = O(k_n)$ is required, where $k_n := n^{\alpha \wedge \delta} I_K$ for MI and I(1) processes. In our simulated sample and also in empirical Section, we observe δ to be around 0.45, so the choice $b = \lceil n^{1/4} \rceil$ does not conflict the rate condition.

5. The $(1 - \alpha)100\%$ percentile interval is $[\hat{G}^{-1}(\alpha/2), \hat{G}^{-1}((1 - \alpha)/2)]$, where $\hat{G}$ is the empirical CDF of $\hat{\beta}_\tau^*$ from step 4.

5.1 Results of Simulation Study 1 - IID Errors

We first examine the size performance under *iid* settings. The test methods are t test, percentile method using MBB technique, IVX-QR, IVX-QR-HAC, and MBB IVX-QR. Each test is conducted at various quantiles with varying degrees of persistent predictors. In particular, we compare the size of MBB IVX-QR to those of IVX-QR and IVX-QR-HAC at the upper and lower quantiles. Although all these tests are designed to have robust size performance across quantiles, our results suggest that MBB IVX-QR significantly outperforms IVX-QR and IVX-QR-HAC at the tails.

Figure 1 shows the results. We summarize the results as follows. First, among these five methods, t test has the largest size distortion regardless of I(1) ($c = 0$), MI ($c = -2, -5, -7$) or I(0) ($c = -70$) predictor. This confirms the theory in Lee (2016) that asymptotic distribution of the standard QR t-statistics is nonstandard. It suffers from distortion depending on the persistence of predictor and nonlinear dependence. In our simulation for both $n = 200$ and 700 , we observe that as $|c|$ approaches 0, the distortion of t test grows, reaching its highest level around 20%. Second, our results illustrate that IVX-QR technique can help to correct the nonstandard distortion under I(1), MI and I(0) settings. IVX-QR, IVX-QR-HAC, and MBB IVX-QR exhibit very similar results between $\tau = 0.2$ and $\tau = 0.8$. However, by moving towards tails ($\tau = 0.05, 0.1, 0.2, 0.8, 0.9, 0.95$), the test sizes of IVX-QR and IVX-QR-HAC are inflated. For a fixed quantile at the tails, these size distortions significantly increase as $|c|$ decreases. Similar quantitative results of IVX-QR are also observed in Lee (2016). We suspect that under *iid* settings the size distortion of IVX-QR and IVX-QR-HAC at the tails is a result of inaccurate estimation of the sparsity function. This issue often happens at the lower and upper quantiles when the sample size is small (see Koenker (2005, Section 3.9)). The results of the t test also exhibit the same pattern of over-rejections because it involves sparsity function estimation. However, these over-rejections of t test and IVX-QR at the tails are dampened when the sample size increases from 200 to 700. Third, in comparison to t test and IVX-QR, the tests using MBB can be conservative at tails. Based on our results, both MBB and MBB IVX-QR improve over their corresponding non-bootstrap counterparts by reducing more than 30% of size distortion at tails. Lastly, the MBB IVX-QR approach performs well across all quantiles. This confirms that MBB IVX-QR improves over IVX-QR at tail quantiles, even under an *iid*-error scenario.

5.2 Results of Simulation Study 2 - CHE Errors

We now consider the predictive QR scenario with ARCH(1) errors. This scenario is closely related to many empirical practices where CHE asset returns are often observed. According to previous asymptotic theory under CHE, the covariance matrix cannot be easily estimated. Therefore, in this simulation study we illustrate that test results can be adversely affected when the estimation of

the covariance matrix is involved. Moreover, we show MBB IVX-QR is robust across all quantiles, substantially improving inference accuracy over all the other methods.

The results are shown in Figure 2, and summarized here. First, we observe that t tests exhibit a U-shaped empirical rejection frequencies and the worst size distortion among all tests. Although these size patterns under CHE settings are similar to the *iid* settings, they are different in terms of scale. In Figure 2a, the size of t tests reaches 0.3 in most of the cases of c 's. Second, unlike the *iid* settings, the tail performances of t test, IVX-QR, and IVX-QR-HAC do not improve as the sample size increases. This may imply that the negative impact of the sparsity function estimation errors is exacerbated at the tails by the CHE errors. Thus, increasing sample size makes the inaccurate estimation issue worse. Third, due to the different asymptotic distribution under CHE, IVX-QR and IVX-QR-HAC do not perform well even at the median. In the figures, for $n = 200$ and 700 , the smallest sizes of IVX-QR and IVX-QR-HAC are around 0.1 for $\tau = 0.5$. The largest size is 0.215 at $\tau = 0.95$. Finally, MBB IVX-QR has the best size control among all tests. The performance of this test is stable across quantiles for all $I(1)$, MI and $I(0)$ scenarios, confirming the robust property of MBB IVX-QR to CHE effects.

In the Online Appendix, for a robustness check of the size performance over a different block length, we conduct a simulation study for $b = n^{1/5}$. The results of using different block lengths are very similar, showing a considerable size improvement of MBB IVX-QR over the existing methods. The power comparison of MBB IVX-QR to the other methods is also provided. In general, MBB IVX-QR has slightly smaller power than t test and MBB in *iid* case but loses more power in CHE case. On the other hand, MBB IVX-QR is comparable to IVX-QR. In particular, for a local deviation of β from 0, their powers are close to each other. As $|\beta|$ increases, the power of MBB IVX-QR converges to 1 at a slower rate than IVX-QR.

6 Empirical Illustration: Stock Return Quantile Predictability

We illustrate an improved predictive QR inference for stock market index returns (S&P 500) using the MBB IVX-QR approach. The univariate predictive mean regression model

$$Stock_Return(t) = \alpha + \beta \times Economic_Variable(t-1) + \epsilon(t), \quad (18)$$

has been considerably discussed in the economics and finance literature; Goyal and Welch (2003), Campbell and Yogo (2006) and Welch and Goyal (2008), to name a few. However, there is no consensus on which economic variables have significant predictive ability for the mean of stock returns. To provide some different insight, Lee (2016) and Maynard et al. (2011) employed a QR-version of this predictive model. In this Section, we discuss the same predictive QR framework and compare the test results using IVX-QR and MBB IVX-QR.

The same data as in Lee (2016) is employed, ranging from 1927 to 2005⁹. Excess stock returns

⁹The data is available from: <https://sites.google.com/site/jihyung412/research>.

(called $Stock_Return(t)$) are calculated as:

$$Stock_Return(t) = \log \left(\frac{P(t) + D(t)}{P(t-1)} \right) - \log(Rfree(t) + 1),$$

where $P(t)$ and $D(t)$ indicate the S&P 500 index and dividends at time t , respectively. $Rfree(t)$ is the 1-month treasury bill rate at time t . The eight persistent predictors are dividend price ratio (dp), dividend payout ratio (de), earnings price ratio (ep), book to market ratio (bm), net equity expansion ($ntis$), treasury bills (tbl), term spread (tms), and default yield spread (dfy). The definition of these variables follows Welch and Goyal (2008). Figure 3 plots these variables from January 1927 to December 2005, signifying highly persistent patterns over time.

As shown in Table 1, inference results of predictors using IVX-QR and MBB IVX-QR are different for some variables at certain quantiles. The results shown with "*" imply the rejection of the null hypothesis of no predictability at 5% level. We observe that at several lower and upper tail quantiles, IVX-QR tends to reject the null hypothesis more often than MBB IVX-QR. For example, when dividend payout ratio (de) is the predictor, p-value of IVX-QR is smaller than 0.05 when $\tau = 0.05, 0.1, 0.2, 0.3, 0.4$ and $\tau = 0.8, 0.9, 0.95$, while 95% confidence intervals of MBB IVX-QR reject none of the null hypothesis $\beta_\tau = 0$ at all quantiles. The similar conflicting results can also be found, for example, at dp when $\tau = 0.05, 0.2, 0.7, 0.95$, bm when $\tau = 0.05, 0.1, 0.2, 0.3, 0.9, 0.95$.

The contradictory results between IVX-QR and MBB IVX-QR at lower and upper tail quantiles may indicate some level of over-rejections using IVX-QR, which was already conjectured from the simulation results in Lee (2016) and Section 5 above. Under both *iid* and CHE errors, MBB IVX-QR can have better size controls especially at lower and upper tail quantiles. In finding significant economic and financial variables to predict excess stock return quantiles, MBB IVX-QR can be a safer inferential tool for applied researchers since it provides a more conservative test.

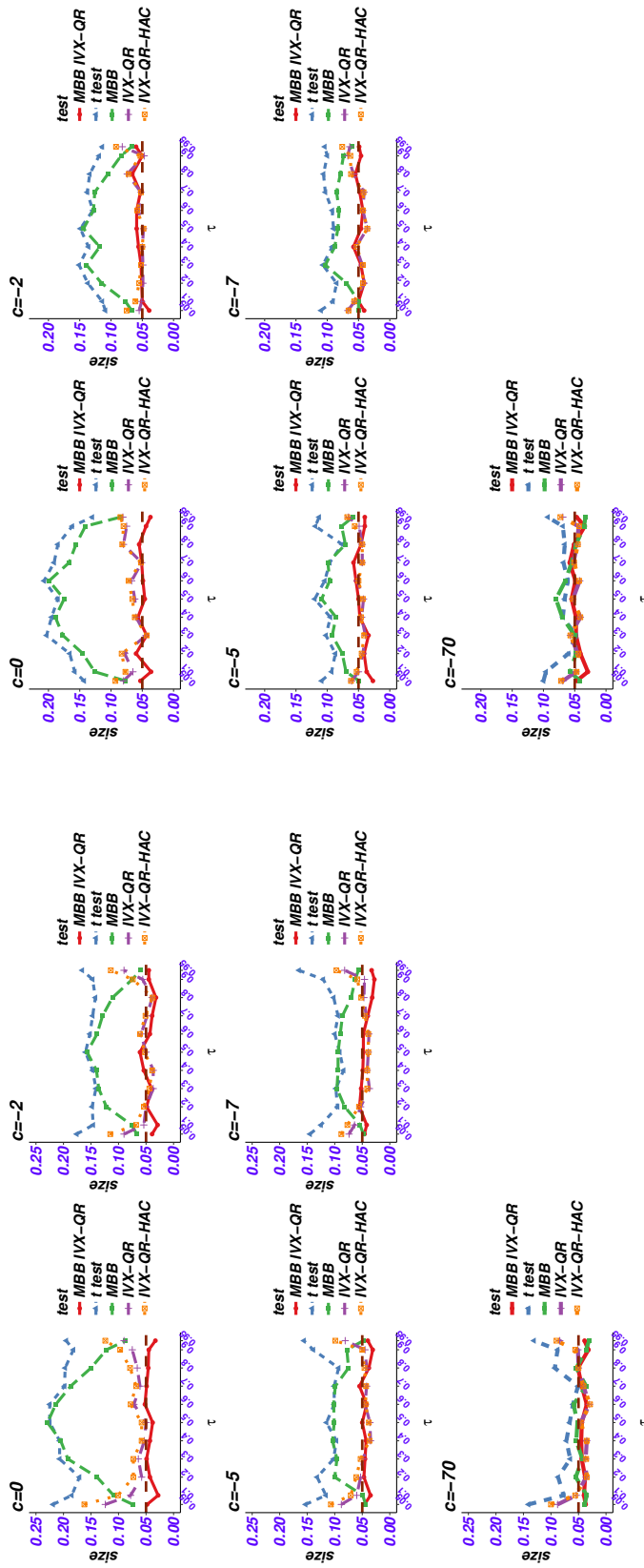
7 Summary and Concluding Remarks

Stock return predictability has been an important topic in economics and finance but the empirical conclusion is still controversial. Meanwhile, valid econometric inferential methods have been carefully developed in the predictive mean and quantile regression literature.

In this paper, we study the predictive quantile regression models with a particular emphasis on two important stylized facts; the predictor persistence and the conditional heteroskedasticity. A valid and easy-to-use inference procedure is proposed and labeled as MBB IVX-QR. As its name indicates, the main development is to combine two techniques, (i) IVX filtering and (ii) the moving block bootstrap. In essence, IVX-filtering removes the nonstationary distortion, while the block bootstrap accommodates the conditional heteroskedasticity. Simulation and empirical results confirm the benefit of the new methods, guarding against the type-I errors arising from persistence and conditional heteroskedasticity, two universal stylized facts in financial data. Therefore MBB IVX-QR is well-suited to empirical exercises in the predictive regression literature.

Acknowledgement

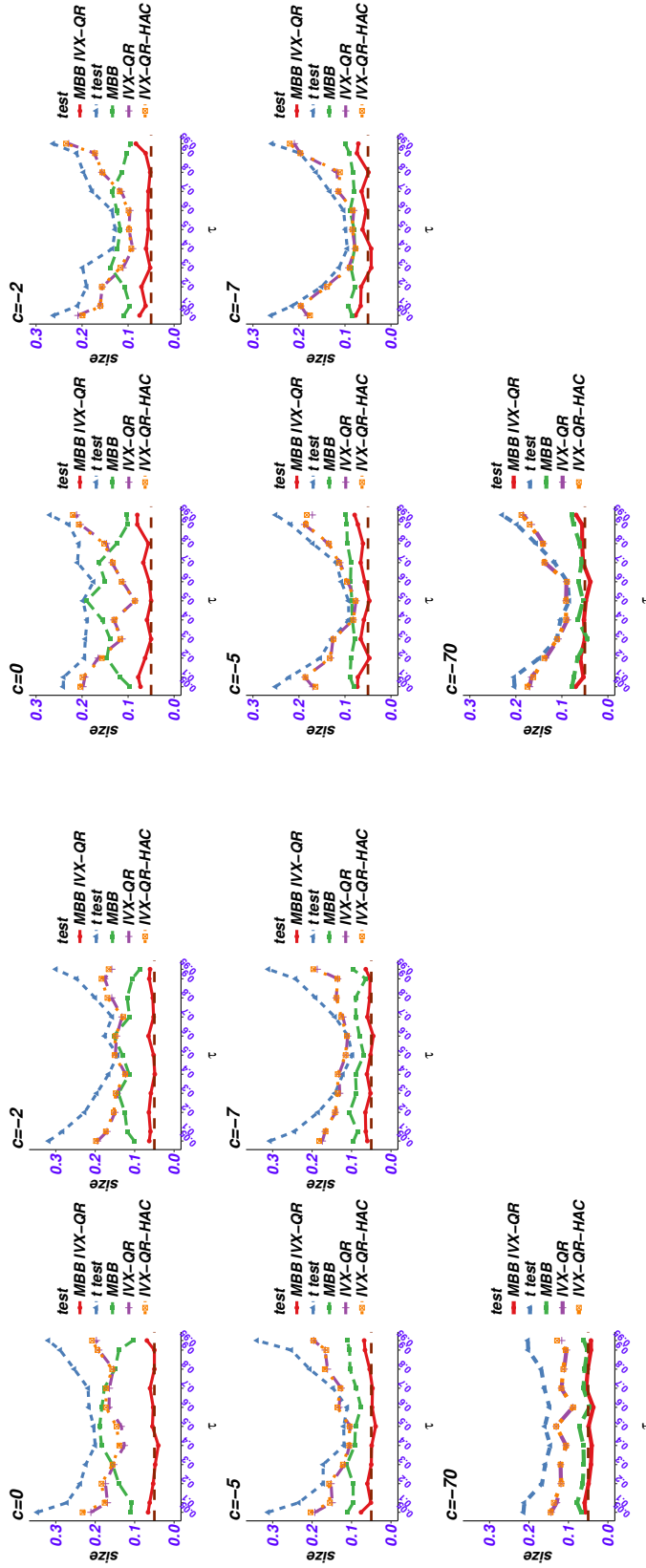
This paper is based on Rui Fan's Ph.D. dissertation at the University of Illinois at Urbana-Champaign. We thank the Committee members Anil Bera, Roger Koenker and Xiaofeng Shao who provided insight and expertise that greatly improved the research. We would also like to show our gratitude to Dan Bernhardt, Zongwu Cai, Thomas Fomby, Antonio Galvao, Sílvia Gonçalves, Daniel Millimet, and seminar participants at UIUC, University of Iowa, Seoul National University, Peking University, Southern Methodist University, Binghamton University and Rensselaer Polytechnic Institute. We are also immensely grateful to two anonymous referees and editor for their constructive comments and suggestions.



(a) Small sample: $N = 200$

(b) Large sample: $N = 700$

Figure 1: Size performance: I.I.D. errors



(a) Small sample: $N = 200$

(b) Large sample: $N = 700$

Figure 2: Size performance: CHE errors

Figure 3: Predictors: 1927:01 - 2005:12

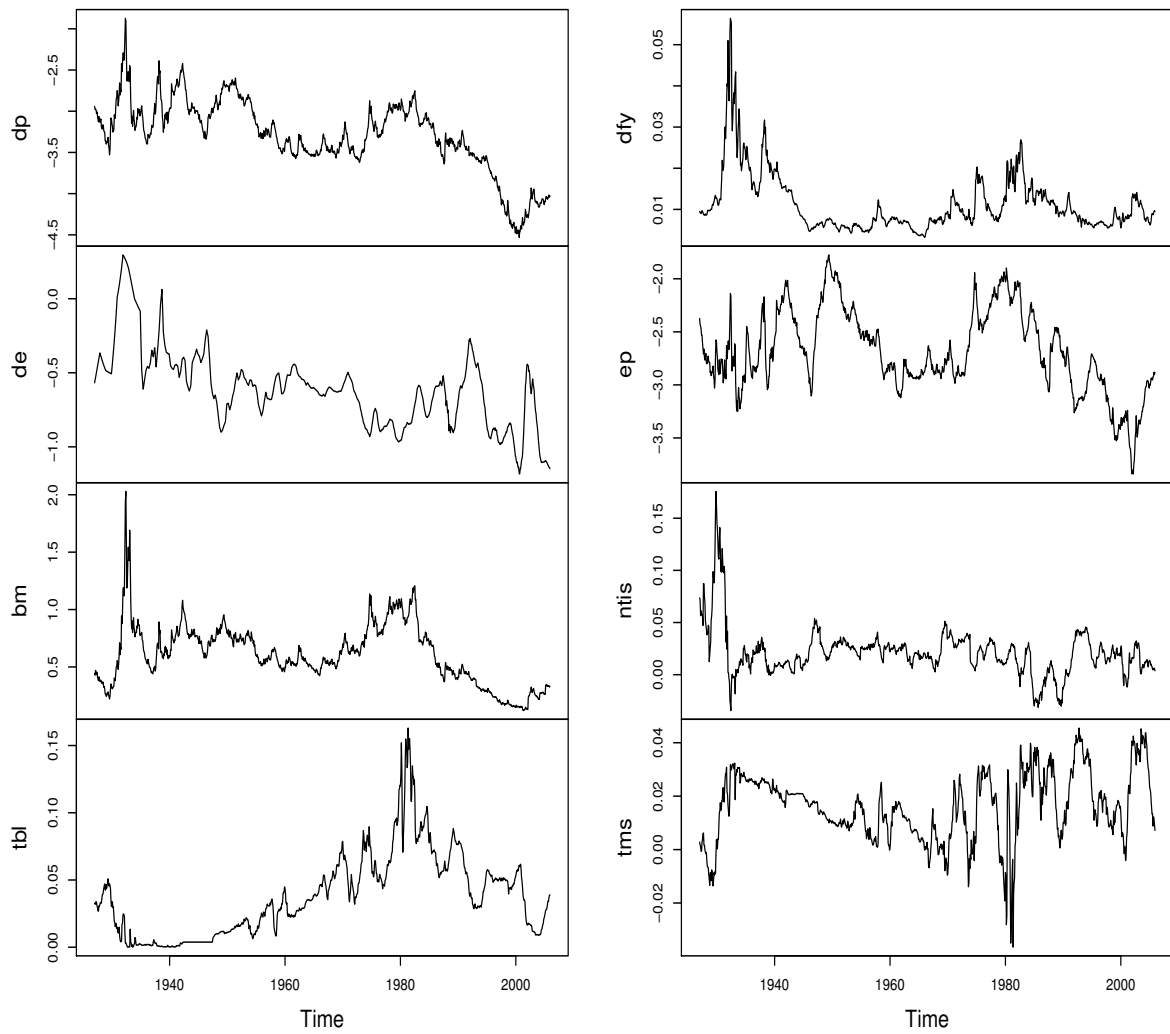


Table 1: Tests for stock returns (S&P500) quantile predictability: 1927:01 - 2005:12

τ	dp		de		bm		tbl	
	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR
0.05	0.003*	(-0.125, 0.006)	0.000*	(-0.137, 0.007)	0.001*	(-0.252, 0.030)	0.604	(-0.598, 0.301)
0.1	0.000*	(-0.131, -0.000)*	0.000*	(-0.116, 0.004)	0.000*	(-0.278, 0.004)	0.689	(-0.467, 0.206)
0.2	0.002*	(-0.089, 0.001)	0.000*	(-0.073, 0.016)	0.001*	(-0.239, 0.007)	0.060	(-0.334, 0.005)
0.3	0.123	(-0.072, -0.003)*	0.000*	(-0.045, 0.005)	0.005*	(-0.146, 0.008)	0.006*	(-0.372, 0.023)
0.4	0.749	(-0.052, -0.000)*	0.015*	(-0.039, 0.011)	0.511	(-0.093, 0.025)	0.003*	(-0.355, -0.055)*
0.5	0.605	(-0.039, 0.012)	0.375	(-0.035, 0.007)	0.792	(-0.065, 0.045)	0.011*	(-0.282, -0.022)*
0.6	0.157	(-0.022, 0.022)	0.534	(-0.026, 0.005)	0.125	(-0.027, 0.088)	0.007*	(-0.351, 0.036)
0.7	0.004*	(-0.018, 0.033)	0.096	(-0.034, 0.007)	0.001*	(0.002, 0.097)*	0.097	(-0.328, 0.052)
0.8	0.059	(-0.015, 0.033)	0.027*	(-0.030, 0.012)	0.023*	(0.008, 0.118)*	0.016*	(-0.249, 0.054)
0.9	0.057	(-0.021, 0.054)	0.000*	(-0.034, 0.038)	0.009*	(-0.014, 0.216)	0.001*	(-0.371, 0.13)
0.95	0.004*	(-0.035, 0.074)	0.000*	(-0.016, 0.086)	0.008*	(-0.001, 0.214)	0.146	(-0.501, 0.381)

τ	dfy		ep		ntis		tms	
	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR	IVX-QR	MBB IVX-QR
0.05	0.000*	(-6.179, -3.229)*	0.917	(-0.084, 0.035)	0.056	(-0.887, 0.470)	0.175	(-1.746, 0.140)
0.1	0.000*	(-4.489, -2.063)*	0.903	(-0.063, 0.045)	0.003*	(-0.701, 0.281)	0.063	(-1.181, 0.483)
0.2	0.000*	(-3.434, -0.802)*	0.579	(-0.038, 0.014)	0.003*	(-0.576, 0.143)	0.219	(-0.626, 0.471)
0.3	0.000*	(-2.325, -0.296)*	0.759	(-0.04, 0.012)	0.270	(-0.433, 0.074)	0.612	(-0.358, 0.346)
0.4	0.030*	(-1.691, 0.272)	0.801	(-0.035, 0.019)	0.186	(-0.267, 0.117)	0.229	(-0.230, 0.385)
0.5	0.747	(-0.763, 0.82)	0.821	(-0.027, 0.018)	0.826	(-0.255, 0.106)	0.352	(-0.130, 0.334)
0.6	0.025*	(-0.319, 1.193)	0.975	(-0.021, 0.019)	0.741	(-0.300, 0.060)	0.242	(-0.210, 0.383)
0.7	0.000*	(0.118, 1.957)*	0.534	(-0.015, 0.021)	0.582	(-0.218, 0.157)	0.703	(-0.227, 0.507)
0.8	0.000*	(0.44, 2.879)*	0.500	(-0.018, 0.027)	0.834	(-0.237, 0.124)	0.170	(-0.029, 0.566)
0.9	0.000*	(1.514, 4.463)*	0.288	(-0.012, 0.044)	0.934	(-0.271, 0.244)	0.000*	(0.219, 1.169)*
0.95	0.000*	(1.968, 5.047)*	0.260	(-0.029, 0.042)	0.966	(-0.462, 0.321)	0.000*	(0.354, 1.707)*

A Technical Appendix

A.1 Supporting Lemmas

I(0) x_t case is covered by Fitzenberger (1997, Theorem 3.3) so in this Section, we provide MBB lemmas for MI-I(1) cases where the IVX variables are less persistent than the predictors, i.e., $\delta < \min(\alpha, 1)$. All the proofs for the supporting lemmas are provided in the online Appendix.

Let $\tilde{Y}_i^\dagger = \frac{1}{\sqrt{b}} \sum_{j=i}^{i+b-1} m_j^{*\dagger}$ denote the normalized average of the i th resampled block under the MBB, and $B_i^\dagger = \frac{1}{\sqrt{b}} \sum_{t=i}^{i+b-1} m_t^\dagger$ denote the normalized average of the block $(m_i^\dagger, \dots, m_{i+b-1}^\dagger)$, $i \geq 1$. Each of the normalized block sums, after the IVX-filtration, will satisfy CLT as $b \rightarrow \infty$. However, to provide MBB consistency, the cross-product of some higher moments should be reasonably bounded (see the proof of Lemma A.4). One sufficient condition for this is $b = O(k_n)$, which is given in Assumption 4.1.

Lemma A.1 *The second moment of the normalized instruments, $Z_t^\dagger$, is bounded for all t :*

$$E(Z_t^{\dagger 2}) = O(1). \quad (19)$$

Lemma A.2 *Let $q = n - b + 1$ and $q = O(n)$, then*

$$\frac{1}{q} \sum_{i=1}^q \frac{1}{b} \sum_{j=i}^{i+b-1} m_j^\dagger = \frac{1}{n} \sum_{t=1}^n m_t^\dagger + O_p\left(\frac{\sqrt{b}}{n}\right). \quad (20)$$

Corollary A.1 *Let $\bar{m}_n^\dagger := n^{-1} \sum_{t=1}^n m_t^\dagger$. From Lemma A.2, if $P^*(\tilde{Y}_1^\dagger = B_j^\dagger) = 1/q$, $1 \leq j \leq q$,*

$$E^* \tilde{Y}_1^\dagger = \frac{1}{q} \sum_{i=1}^q B_i^\dagger = \frac{1}{q} \sum_{i=1}^q \frac{1}{\sqrt{b}} \sum_{t=i}^{i+b-1} m_t^\dagger = \sqrt{b} \bar{m}_n^\dagger + O_p\left(\frac{b}{n}\right). \quad (21)$$

Lemma A.3 *Let $z_t := \sum_{i=1}^t R_{nz}^{t-i} u_{xi}$ be a mildly integrated instrument that satisfies*

$$z_t = R_{nz} z_{t-1} + u_{xt}, \quad t \in \{1, \dots, n\}, z_0 = 0.$$

Suppose Assumption 3.1 holds, for $1 \leq j \leq b$, $1 \leq t \leq n$,

1. $E(z_{t+j}^2 | \mathcal{F}_t) \leq \sum_{i,h=1}^t R_{nz}^{(t+j-i)+(t+j-h)} u_{xi} u_{xh} + O_p(b).$
2. $E(z_{t+j} \Psi_{nt+j} | \mathcal{F}_t) \leq \sum_{i=1}^t \sum_{h=1}^{t+1} \sum_{k=0}^{h-1} R_{nz}^{(t+j-i)+(t+j-h)} R_n^{h-1-k} u_{xi} u_{xk} + O_p(b^2).$
3. $E(\Psi_{nt+j}^2 | \mathcal{F}_t) \leq \sum_{i,h=1}^{t+1} \sum_{f=0}^{i-1} \sum_{k=0}^{h-1} R_{nz}^{(t+j-i)+(t+j-h)} R_n^{(i-1-f)+(h-1-k)} u_{xf} u_{xk} + O_p(b^3).$

Lemma A.4 Under Assumption 2.1, 3.1 and 4.1,,

$$\frac{1}{q} \sum_{i=1}^q B_i^{\dagger 2} \rightarrow_p \tau(1 - \tau)V_{cxz}. \quad (22)$$

Lemma A.5 The long-run variance of $m_t^\dagger$ satisfies:

$$(1/n) \sum_{s,t=1}^n \text{Cov}(m_t^\dagger, m_s^\dagger) = O_p(1). \quad (23)$$

Definition A.1 (Uniform Integrability) (Serfling (1980, page 13))

A sequence of random variable $\{Y_n\}$ is uniformly integrable if

$$\lim_{c \rightarrow \infty} \sup_n E\{|Y_n| \mathbf{1}(|Y_n| > c)\} = 0.$$

Lemma A.6 (Lemma B of Serfling (1980, page 15))

Suppose that $X_n \rightarrow_d X$ and $\lim_n E|X_n|^r = E|X|^r < \infty$ for $r > 0$. Then the sequence $\{X_n^r\}$ is uniformly integrable.

Lemma A.7 Define the QR estimator $\tilde{\beta}_\tau$ under P^* conditional on data following Gregory, Lahiri and Nordman (2018, GLN hereafter) as:

$$\tilde{\beta}_\tau = \arg \min_{\beta} \sum_{t=1}^{\ell} E^* \rho_\tau(y_t^* - \beta' z_{t-1}^*).$$

Then,

$$(\ell k_n)^{1/2} \left(\hat{\beta}_\tau^{IVXQR} - \tilde{\beta}_\tau \right) = o_p(1). \quad (24)$$

A.2 Proofs for Main Theorems

Proof of Theorem 3.1. For $I(0)$ x_t , we have

$$\sqrt{n} \left(\hat{\beta}_\tau^{QR} - \beta_\tau \right) = \Omega_1^{-1} \frac{1}{\sqrt{n}} \sum_{t=1}^n \psi_\tau(u_{t\tau}) X_{t-1} + o_p(1),$$

where

$$\Omega_1 = f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{X_{t-1} X_{t-1}'}{\sigma_t} \right],$$

and, by MGCLT,

$$\frac{1}{\sqrt{n}} \sum_{t=1}^n \psi_{\tau}(u_{t\tau})x_{t-1} \Rightarrow N(0, \tau(1-\tau) E[X_{t-1}X'_{t-1}]).$$

Thus

$$\sqrt{n}(\hat{\beta}_{\tau}^{QR} - \beta_{\tau}) \Rightarrow N\left(0, \frac{\tau(1-\tau)}{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))^2} E\left[\frac{X_{t-1}X'_{t-1}}{\sigma_t}\right]^{-1} E[X_{t-1}X'_{t-1}] E\left[\frac{X_{t-1}X'_{t-1}}{\sigma_t}\right]^{-1}\right).$$

In a similar way, for (MI) x_t

$$\sqrt{nk_n}(\hat{\beta}_{\tau}^{QR} - \beta_{\tau}) \Rightarrow N\left(0, \frac{\tau(1-\tau)}{f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau))^2} \tilde{V}_{xx}^{-1} V_{xx} \tilde{V}_{xx}^{-1}\right)$$

with $V_{xx} = \int_0^{\infty} e^{rC} \Omega_{xx} e^{rC} dr$ and $\tilde{V}_{xx} = E\left[\frac{1}{\sigma_t}\right] V_{xx}$. See also Kostakis et al (2014, Theorem B.4).

I(1) case is already discussed in Section 3.1, so is omitted. ■

Proof of Theorem 3.2. Following the proof of Lee (2016) but using CHE innovation specification we have

$$(nk_n)^{1/2}(\hat{\beta}_{1\tau}^{IVXQR} - \beta_{1,\tau}) = (\tilde{M}_{\beta_{\tau},n})^{-1} G_{\tau,n} + o_p(1),$$

where

$$\begin{aligned} \tilde{M}_{\beta_{\tau},n} &:= \sum_{t=1}^n \tilde{f}_{u_{t\tau},t-1}(0) \tilde{Z}_{t-1,n} \tilde{Z}'_{t-1,n}, \\ \tilde{f}_{u_{t\tau},t-1}(0) &= f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau)) \frac{1}{\sigma_t}, \end{aligned}$$

hence

$$\tilde{M}_{\beta_{\tau},n} \rightarrow_p \tilde{M}_{\beta_{\tau}} \equiv \begin{cases} f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau)) E\left[\frac{X_{t-1}X'_{t-1}}{\sigma_t}\right], & \text{for (I0),} \\ f_{\varepsilon}(F_{\varepsilon}^{-1}(\tau)) E\left[\frac{1}{\sigma_t}\right] V_{cxz}. & \text{for (MI) and (I1),} \end{cases}$$

Moreover,

$$G_{\tau,n} := \sum_{t=1}^n \tilde{Z}_{t-1,n} \psi_{\tau}(u_{t\tau}) \Rightarrow G_{\tau} \equiv \begin{cases} N(0, \tau(1-\tau) E[X_{t-1}X'_{t-1}]) & \text{for (I0),} \\ N(0, \tau(1-\tau) V_{cxz}) & \text{for (MI) and (I1),} \end{cases}.$$

Thus, the desired result follows. ■

Theorem A.2 (Stability Condition) Let $\bar{m}_{\ell}^{*\dagger}$ denote the average of resampled data under the MBB, $\bar{m}_{\ell}^{*\dagger} = (1/\ell) \sum_{t=1}^{\ell} m_t^{*\dagger}$. Under the assumptions 2.1, 3.1 and 4.1,

$$\ell Var^*(\bar{m}_{\ell}^{*\dagger}) \rightarrow_p \tau(1-\tau) V_{cxz}, \quad \text{as } n \rightarrow \infty. \quad (25)$$

Proof of Theorem A.2. Using the fact that $\tilde{Y}_i^\dagger$ are *iid* under P^* , we can observe

$$\ell Var^*(\bar{m}_\ell^{*\dagger}) = \ell Var^*\left(\frac{1}{\sqrt{\ell m}} \sum_{i=1}^m \tilde{Y}_i^\dagger\right) = Var^*(\tilde{Y}_1^\dagger).$$

By Corollary A.1 and the fact that $P^*(\tilde{Y}_1^\dagger = B_j^\dagger) = 1/q$, $1 \leq j \leq q$,

$$Var^*(\tilde{Y}_1^\dagger) = \frac{1}{q} \sum_{i=1}^q (B_i^\dagger - E^* \tilde{Y}_1^\dagger)^2 \quad (26)$$

$$= \frac{1}{q} \sum_{i=1}^q \left\{ B_i^\dagger - \sqrt{b} \bar{m}_n^\dagger - O_p\left(\frac{b}{n}\right) \right\}^2 \quad (27)$$

$$= \frac{1}{q} \sum_{i=1}^q \left(B_i^\dagger - \sqrt{b} \bar{m}_n^\dagger \right)^2 - \frac{2}{q} \sum_{i=1}^q \left(B_i^\dagger - \sqrt{b} \bar{m}_n^\dagger \right) O_p\left(\frac{b}{n}\right) + O_p\left(\frac{b}{n}\right)^2 \quad (28)$$

Providing that $\sqrt{b} \bar{m}_n^\dagger = \sqrt{\frac{b}{n}} \cdot (n^{-1/2} \sum_{t=1}^n m_t^\dagger) = O_p(\sqrt{\frac{b}{n}})$, and that $b = o(n)$,

$$Var^*(\tilde{Y}_1^\dagger) = \frac{1}{q} \sum_{i=1}^q \left(B_i^\dagger - \sqrt{b} \bar{m}_n^\dagger \right)^2 + o_p(1). \quad (29)$$

In order to prove 25, it suffices to show

$$\frac{1}{q} \sum_{i=1}^q \left(B_i^\dagger - b^{1/2} \bar{m}_n^\dagger \right)^2 \rightarrow_p \tau(1 - \tau) V_{cxz}. \quad (30)$$

This can be decomposed further as:

$$\begin{aligned} \frac{1}{q} \sum_{i=1}^q \left(B_i^\dagger - \sqrt{b} \bar{m}_n^\dagger \right)^2 &= \frac{1}{q} \sum_{i=1}^q B_i^{\dagger 2} - \frac{2}{q} \sum_{i=1}^q B_i^\dagger \sqrt{b} \bar{m}_n^\dagger + \frac{1}{q} \sum_{i=1}^q \left(\sqrt{b} \bar{m}_n^\dagger \right)^2 \\ &\equiv A_n - 2C_n + D_n. \end{aligned}$$

Next we will show: (a) $A_n \rightarrow_p \tau(1 - \tau) V_{cxz}$; (b) $C_n \rightarrow_p 0$; (c) $D_n \rightarrow_p 0$.

(a) This result is obtained by Lemma A.4.

(b)

$$\begin{aligned} C_n &= \left(\sqrt{b} \bar{m}_n + O_p\left(\frac{b}{n}\right) \right) \cdot \sqrt{b} \bar{m}_n \\ &= \left[\sqrt{\frac{b}{n}} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^n m_t^\dagger + O_p\left(\frac{b}{n}\right) \right] \cdot \sqrt{\frac{b}{n}} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^n m_t^\dagger \\ &= \left[\frac{b}{n} \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^n m_t^\dagger + O_p\left(\left(\frac{b}{n}\right)^{3/2}\right) \right] \cdot \frac{1}{\sqrt{n}} \sum_{t=1}^n m_t^\dagger \\ &= o_p(1) \cdot O_p(1) \end{aligned}$$

$$= o_p(1).$$

(c)

$$D_n = \frac{1}{q} \sum_{i=1}^q b \bar{m}_n^2 = \frac{b}{n} \left(\frac{1}{\sqrt{n}} \sum_{t=1}^n m_t^\dagger \right)^2 = o_p(1) O_p(1) = o_p(1).$$

■

Proof of Theorem 4.1. Since $\sqrt{n}(\bar{m}_n^\dagger - E m_t^\dagger) \Rightarrow G_\tau$ (Lee (2016), Lemma A.2), where

$$G_\tau = \begin{cases} \mathcal{N}(0, \tau(1-\tau)E[X_{t-1}X'_{t-1}]), & \text{for (I0),} \\ \mathcal{N}(0, \tau(1-\tau)V_{cxz}), & \text{for (MI) and (II),} \end{cases}$$

and G_τ is a continuous distribution, by Polyá's Theorem,

$$\sup_{x \in R} \left| P(\sqrt{n}(\bar{m}_n^\dagger - E m_t^\dagger) \leq x) - G_\tau(x) \right| \rightarrow_p 0, \quad \text{as } n \rightarrow \infty. \quad (31)$$

Hence, it suffices to show that

$$\sup_{x \in R} \left| P^*(\sqrt{\ell}(\bar{m}_\ell^{*\dagger} - \bar{m}_n^\dagger) \leq x) - G_\tau(x) \right| \rightarrow_p 0, \quad \text{as } n \rightarrow \infty. \quad (32)$$

Recall that $\tilde{Y}_i^\dagger = \frac{1}{\sqrt{b}} \sum_{j=i}^{i+b-1} m_t^{*\dagger}$. We can obtain

$$\sqrt{\ell} \bar{m}_\ell^{*\dagger} = \frac{1}{\sqrt{m}} \sum_{i=1}^m \tilde{Y}_i^\dagger. \quad (33)$$

Let $\hat{\Delta}_n(\lambda) := \frac{1}{m} \sum_{i=1}^m E^* \left\{ \left(\tilde{Y}_i^\dagger - E^* \tilde{Y}_i^\dagger \right)^2 \mathbf{1} \left(\left| \tilde{Y}_i^\dagger - E^* \tilde{Y}_i^\dagger \right| > \lambda \sqrt{m} \right) \right\}$, for $\lambda > 0$. Using the fact that $\tilde{Y}_i^\dagger$, $i = 1, 2, \dots, m$ are iid under P^* , and that $P^*(\tilde{Y}_1^\dagger = B_j^\dagger) = 1/q$, $1 \leq j \leq q$, for any $\epsilon > 0$,

$$\begin{aligned} Pr(\hat{\Delta}_n(\lambda) > \epsilon) &\leq \epsilon^{-1} E(\hat{\Delta}_n(\lambda)) \\ &= \epsilon^{-1} E \left\{ E^* \left[\left(\tilde{Y}_1^\dagger - E^* \tilde{Y}_1^\dagger \right)^2 \mathbf{1} \left(\left| \tilde{Y}_1^\dagger - E^* \tilde{Y}_1^\dagger \right| > \lambda \sqrt{m} \right) \right] \right\} \\ &= \epsilon^{-1} E \left\{ \frac{1}{q} \sum_{i=1}^q \left[\left(B_i^\dagger - \frac{1}{q} \sum_{j=1}^q B_j^\dagger \right)^2 \mathbf{1} \left(\left| B_i^\dagger - \frac{1}{q} \sum_{j=1}^q B_j^\dagger \right| > \lambda \sqrt{m} \right) \right] \right\} \\ &= (q\epsilon)^{-1} \sum_{i=1}^q E \left\{ \left[\left(B_i^\dagger - \frac{1}{q} \sum_{j=1}^q B_j^\dagger \right)^2 \mathbf{1} \left(\left| B_i^\dagger - \frac{1}{q} \sum_{j=1}^q B_j^\dagger \right| > \lambda \sqrt{m} \right) \right] \right\} \\ &\leq 4(q\epsilon)^{-1} \sum_{i=1}^q \left[E(B_i^\dagger)^2 \mathbf{1} \left(\left| B_i^\dagger \right| > (\lambda/2) \sqrt{m} \right) + E \left(\frac{1}{q} \sum_{j=1}^q B_j^\dagger \right)^2 \mathbf{1} \left(\left| \frac{1}{q} \sum_{j=1}^q B_j^\dagger \right| > (\lambda/2) \sqrt{m} \right) \right] \\ &\leq 4(q\epsilon)^{-1} \sum_{i=1}^q \left[E(B_i^\dagger)^2 \mathbf{1} \left(\left| B_i^\dagger \right| > (\lambda/2) \sqrt{m} \right) + E \left(\frac{1}{q} \sum_{j=1}^q B_j^\dagger \right)^2 \right] \end{aligned}$$

$$= o(1),$$

since

$$\lim_{n \rightarrow \infty} E \left[(B_i^\dagger)^2 \mathbf{1} \left(|B_i^\dagger| > (\lambda/2)\sqrt{m} \right) \right] = 0, \quad (34)$$

and by Lemma A.2, $b = o(n)$, $q = O(1)$, and $E(m_t^\dagger)^2 = O(1)$, for $1 \leq t \leq n$, $E(m_i^\dagger m_j^\dagger) = 0$, for $i \neq j$, $1 \leq i, j \leq n$,

$$\begin{aligned} E \left(\frac{1}{q} \sum_{j=1}^q B_j^\dagger \right)^2 &= E \left\{ \frac{\sqrt{b}}{q} \sum_{t=1}^n m_t^\dagger - \frac{1}{q\sqrt{b}} \left[\sum_{j=1}^{b-1} (b-j)(m_j^\dagger + m_{n-j+1}^\dagger) \right] \right\}^2 \\ &\leq E \left\{ b \left(\frac{1}{n} \sum_{t=1}^n m_t^\dagger \right)^2 - \frac{2b}{q^2} \sum_{t=1}^n \sum_{j=1}^{b-1} m_t^\dagger (m_j^\dagger + m_{n-j+1}^\dagger) + \frac{b}{q^2} \left[\sum_{j=1}^{b-1} (m_j^\dagger + m_{n-j+1}^\dagger) \right]^2 \right\} \\ &= \frac{b}{n^2} \sum_{t=1}^n E(m_t^\dagger)^2 - \frac{2b}{n^2} \sum_{j=1}^{b-1} E(m_j^\dagger)^2 - \frac{2b}{n^2} \sum_{j=1}^{b-1} E(m_{n-j+1}^\dagger)^2 + \frac{b}{n^2} E \left[\sum_{j=1}^{b-1} (m_j^\dagger + m_{n-j+1}^\dagger) \right]^2 \\ &= O\left(\frac{b}{n}\right) + O\left(\frac{b^2}{n^2}\right) + O\left(\frac{b^2}{n^2}\right) + O\left(\frac{b^2}{n^2}\right) \\ &= o(1). \end{aligned}$$

In order to prove (34), we define the triangular array $\{V_{n,i} : i = 1, \dots, n-b+1\}$ as

$$V_{n,i} = B_i^{\dagger 2} = \frac{1}{b} \left(\sum_{i=1}^{i+b-1} m_t^\dagger \right)^2.$$

For fixed $i = 1, \dots, q$, $E(m_t^\dagger)^2 < \infty$, we observe that

$$\frac{1}{\sqrt{b}} \sum_{i=1}^{i+b-1} m_t^\dagger \Rightarrow \mathcal{N}(0, E(m_t^\dagger)^2),$$

and

$$E \left[\left(\frac{1}{\sqrt{b}} \sum_{i=1}^{i+b-1} m_t^\dagger \right)^2 \right] = E(m_t^\dagger)^2 < \infty.$$

Hence, by Serfling (1980, page 15, Lemma B), the sequences $\{V_{n,i}\}_{n=n_1}^\infty, \dots, \{V_{n,i}\}_{n=n_q}^\infty$ is uniformly integrable, which implies that, for $\lambda > 0$:

$$E \left[V_{n,i} \mathbf{1} \left(V_{n,i} > \left(\frac{\lambda}{2} \right)^2 m \right) \right] = E \left[(B_i^\dagger)^2 \mathbf{1} \left((B_i^\dagger)^2 > \left(\frac{\lambda}{2} \right)^2 m \right) \right] \rightarrow 0, \text{ as } m \rightarrow \infty,$$

Using this result yields

$$\lim_{n \rightarrow \infty} E \left[(B_i^\dagger)^2 \mathbf{1} \left(|B_i^\dagger| > (\lambda/2)\sqrt{m} \right) \right] = 0,$$

where $m = \frac{n}{b} \rightarrow \infty$, as $n \rightarrow \infty$.

Therefore, we conclude that $\hat{\Delta}_n(\lambda) = o_p(1)$, which implies the Lindeberg's condition:

$$\frac{1}{\ell} \sum_{t=1}^{\ell} E^* \left\{ \left(m_t^{*\dagger} - E^* m_t^{*\dagger} \right)^2 \mathbf{1} \left(\left| m_t^{*\dagger} - E^* m_t^{*\dagger} \right| > \lambda \sqrt{\ell} \right) \right\} = o_p(1), \quad \text{as } n \rightarrow \infty. \quad (35)$$

Next, by Lindeberg's Central Limit Theorem, using Theorem A.2, along with the Lindeberg's condition (35), yields that the conditional distribution of $\sqrt{\ell}(\bar{m}_\ell^{*\dagger} - \bar{m}_n^\dagger)$ given $(m_1^\dagger, \dots, m_n^\dagger)$ converges to G_τ , as $n \rightarrow \infty$. Hence, by *Polya's* Theorem, (32) follows. ■

Proof of Theorem 4.2. We combine the proof of Lee (2016) and some argument from GLN (2018). Recall that the QR estimator $\tilde{\beta}_\tau$ under P^* conditional on data as in Lemma A.7:

$$\tilde{\beta}_\tau = \arg \min_{\beta} \sum_{t=1}^{\ell} E^* \rho_\tau (y_t^* - \beta' \tilde{z}_{t-1}^*),$$

and let $u_{t\tau}^* := y_t^* - \tilde{z}_{t-1}^{*\prime} \tilde{\beta}_\tau$, then $E^* \psi_\tau(u_{t\tau}^*) = 0$ by construction.

Recall that (we use $\hat{\beta}_\tau^*$ instead of $\hat{\beta}_\tau^{IVXQR*}$ within this proof for exposition)

$$\hat{\beta}_\tau^* = \arg \min_{\beta} \sum_{t=1}^{\ell} \rho_\tau (y_t^* - \beta' \tilde{z}_{t-1}^*).$$

We obtain the approximate FOC (Koenker and Bassett, 1982):

$$\ell^{-1/2} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} [\tau - \mathbf{1}(y_t^* \leq \tilde{z}_{t-1}^{*\prime} \hat{\beta}_\tau^*)] = \ell^{-1/2} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} [\psi_\tau(y_t^* - \tilde{z}_{t-1}^{*\prime} \hat{\beta}_\tau^*)] = o_p(1), \quad (36)$$

and

$$\ell^{-1/2} \sum_{t=1}^{\ell} E^* Z_{t-1}^{*\dagger} [\tau - \mathbf{1}(y_t^* \leq \tilde{z}_{t-1}^{*\prime} \tilde{\beta}_\tau)] = \ell^{-1/2} \sum_{t=1}^{\ell} E^* Z_{t-1}^{*\dagger} [\psi_\tau(y_t^* - \tilde{z}_{t-1}^{*\prime} \tilde{\beta}_\tau)] = o_p(1),$$

respectively. Following the proof of Lee (2016, Theorem 3.1), we have

$$\begin{aligned} o_p(1) &= \ell^{-1/2} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) \\ &= \ell^{-1/2} \sum_{t=1}^{\ell} \left\{ \left[Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) - E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) \right] \right. \\ &\quad + \left[E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) - E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) \right] \\ &\quad - \left[Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) - Z_{t-1}^{*\dagger} E^* \psi_\tau(u_{t\tau}^*) \right] - \left[Z_{t-1}^{*\dagger} E^* \psi_\tau(u_{t\tau}^*) - Z_{t-1}^{*\dagger} E_{t-1} E^* \psi_\tau(u_{t\tau}^*) \right] \Big\} \\ &\quad + \ell^{-1/2} \sum_{t=1}^{\ell} \left\{ E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*\prime} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau)) \right. \end{aligned} \quad (37)$$

$$\left. - Z_{t-1}^{*\dagger} E_{t-1}^* \psi_\tau(u_{t\tau}^*) + Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) \right\} \quad (38)$$

$$= \ell^{1/2} \tilde{\lambda}_\ell(\hat{\beta}_\tau^*) + \ell^{-1/2} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + o_p(1), \quad (39)$$

where

$$\tilde{\lambda}_\ell(\hat{\beta}_\tau^*) := \ell^{-1} \sum_{t=1}^{\ell} E_{t-1}^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*'}(\hat{\beta}_\tau^* - \tilde{\beta}_\tau)),$$

and $o_p(1)$ terms are smaller orders than (39) under either P or P^* , see GLN (Lemma 2-(vi) and Lemma 3-(vi), 2018) and Lee (2016). Please also see the proof of Theorems 1-2 (equations (8.3) and (8.5)) in GLN. These additional rounds of approximations in order to avoid the non-differentiability under E^* is from GLN, which we modify into our proof. For more details, see the proof of Lemma A.7 in the online Appendix.

By the Taylor expansion of $\tilde{\lambda}_\ell(\hat{\beta}_\tau^*)$ around $\tilde{\beta}_\tau$:

$$\begin{aligned} \tilde{\lambda}_\ell(\hat{\beta}_\tau^*) &= \tilde{\lambda}_\ell(\tilde{\beta}_\tau) + \left. \frac{\partial \tilde{\lambda}_\ell(\beta_\tau)}{\partial \beta_\tau} \right|_{\tilde{\beta}_\tau} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) + o_p(\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \\ &= \ell^{-1} \sum_{t=1}^{\ell} E_{t-1}^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + \left. \frac{\partial \tilde{\lambda}_\ell(\beta_\tau)}{\partial \beta_\tau} \right|_{\tilde{\beta}_\tau} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) + o_p(\hat{\beta}_\tau^* - \tilde{\beta}_\tau). \end{aligned}$$

Thus, (39) can be written as

$$\ell^{1/2} \left[\ell^{-1} \sum_{t=1}^{\ell} E_{t-1}^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + \left. \frac{\partial \tilde{\lambda}_\ell(\beta)}{\partial \beta} \right|_{\tilde{\beta}_\tau} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) + o_p(\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \right] + \ell^{-1/2} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) = o_p(1). \quad (40)$$

Using our MBB lemmas,

$$\begin{aligned} \tilde{\lambda}_\ell(\beta) &= \ell^{-1} \sum_{t=1}^{\ell} E_{t-1}^* \left[Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*'}(\beta - \tilde{\beta}_\tau)) \right] \\ &= E_{t-1}^* \left[\ell^{-1} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^* - \tilde{z}_{t-1}^{*'}(\beta - \tilde{\beta}_\tau)) \right] \\ &= E_{t-1}^* \left[\frac{1}{mb} \sum_{i=1}^m \sum_{j=1}^{i+b-1} Z_{j-1}^{*\dagger} \psi_\tau(u_{j\tau}^* - \tilde{z}_{j-1}^{*'}(\beta - \tilde{\beta}_\tau)) \right] \\ &= E_{t-1}^* \frac{1}{m} \frac{1}{\sqrt{b}} \sum_{i=1}^m E^* \left[\frac{1}{\sqrt{b}} \sum_{j=1}^{i+b-1} m_j^{*\dagger}(\beta) \right] \\ &= E_{t-1}^* \frac{1}{m} \frac{1}{\sqrt{b}} \sum_{i=1}^m \left[\sqrt{b} \bar{m}_n(\beta) + O_p(b/n) \right] \quad (\text{by Corollary A.1}) \end{aligned}$$

$$\begin{aligned}
&= E_{t-1} \left[n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger \psi_\tau(u_{t\tau} - \tilde{z}'_{t-1}(\beta - \tilde{\beta}_\tau)) \right] + O_p(\sqrt{b}/n) \\
&= n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger E_{t-1} \psi_\tau(u_{t\tau} - \tilde{z}'_{t-1}(\beta - \tilde{\beta}_\tau)) + O_p(\sqrt{b}/n).
\end{aligned}$$

So the partial derivative of $\tilde{\lambda}_\ell(\beta)$ evaluated at $\tilde{\beta}_\tau$ is

$$\left. \frac{\partial \tilde{\lambda}_\ell(\beta)}{\partial \beta} \right|_{\tilde{\beta}_\tau} = -k_n^{1/2} n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0),$$

where $\tilde{f}_{u_{t\tau}, t-1}(0) = f_\varepsilon(F_\varepsilon^{-1}(\tau)) \frac{1}{\sigma_t}$.

Plug this result in (40), we have

$$\begin{aligned}
o_p(1) &= \ell^{1/2} \left[\ell^{-1} \sum_{t=1}^\ell E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) - k_n^{1/2} n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0) (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) + o_p(\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \right] \\
&\quad + \ell^{-1/2} \sum_{t=1}^\ell Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) \\
&= -(\ell k_n)^{1/2} n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0) (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \\
&\quad + \ell^{1/2} o_p(\hat{\beta}_\tau^* - \tilde{\beta}_\tau) + \ell^{-1/2} \sum_{t=1}^\ell Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + \ell^{-1/2} \sum_{t=1}^\ell E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) \\
&= -(\ell k_n)^{1/2} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \cdot n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0) + (\ell k_n)^{1/2} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) o_p((k_n)^{-1/2}) \\
&\quad + \ell^{1/2} \ell^{-1} \sum_{t=1}^\ell Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + \ell^{-1/2} \sum_{t=1}^\ell E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*).
\end{aligned}$$

Then

$$\begin{aligned}
&(\ell k_n)^{1/2} (\hat{\beta}_\tau^* - \tilde{\beta}_\tau) \cdot \left(n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0) + o_p((k_n)^{-1/2}) \right) \\
&= \ell^{1/2} \left[\frac{1}{\ell} \sum_{t=1}^\ell Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) + \frac{1}{\ell} \sum_{t=1}^\ell E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_\tau(u_{t\tau}^*) \right],
\end{aligned}$$

and we know,

$$n^{-1} \sum_{t=1}^n Z_{t-1}^\dagger Z_{t-1}^{\dagger'} \tilde{f}_{u_{t\tau}, t-1}(0) \rightarrow_p \tilde{M}_{\beta_\tau} \equiv \begin{cases} f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{X_{t-1} X'_{t-1}}{\sigma_t} \right], & \text{for (I0)} \\ f_\varepsilon(F_\varepsilon^{-1}(\tau)) E \left[\frac{1}{\sigma_t} \right] V_{cxz}, & \text{for (MI) and (II)}. \end{cases}$$

Note that

$$\begin{aligned}
\frac{1}{\ell} \sum_{t=1}^{\ell} E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_{\tau}(u_{t\tau}^*) &= E_{t-1} E^* \frac{1}{\ell} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_{\tau}(u_{t\tau}^*) \\
&= E_{t-1} E^* \frac{1}{mb} \sum_{i=1}^m \sum_{j=1}^{i+b-1} Z_{j-1}^{*\dagger} \psi_{\tau}(u_{j\tau}^*) \\
&= E_{t-1} \frac{1}{m} \frac{1}{\sqrt{b}} \sum_{i=1}^m \left[\sqrt{b} \bar{m}_n \right] \\
&= n^{-1} \sum_{t=1}^n Z_{t-1}^{\dagger} E_{t-1} \psi_{\tau}(u_{t\tau}),
\end{aligned}$$

and from Theorem 4.1,

$$\ell^{1/2} \left[\frac{1}{\ell} \sum_{t=1}^{\ell} Z_{t-1}^{*\dagger} \psi_{\tau}(u_{t\tau}^*) + \frac{1}{\ell} \sum_{t=1}^{\ell} E_{t-1} E^* Z_{t-1}^{*\dagger} \psi_{\tau}(u_{t\tau}^*) \right] \Rightarrow G_{\tau} = \begin{cases} \mathcal{N}(0, \tau(1-\tau)E[X_{t-1}X'_{t-1}]), & \text{for (I0),} \\ \mathcal{N}(0, \tau(1-\tau)V_{cxz}), & \text{for (MI) and (II).} \end{cases}$$

Theorem 4.2 now follows, together with $\sqrt{\ell k_n}(\tilde{\beta}_{\tau} - \hat{\beta}_{\tau}^{IVXQR}) = o_p(1)$, from Lemma A.7.

■

References

- [1] L. I. Boneva, D. Kendall, and I. Stefanov. Spline transformations: Three new diagnostic aids for the statistical data-analyst. *Journal of the Royal Statistical Society. Series B (Methodological)*, 33(1):1–71, 1971.
- [2] Z. W. Cai, X. S. Liao, and H. Q. Chen. A weighted variable addition approach to test predictability in quantile regression with persistent predictors. *Xiamen University, Working Paper*, 2017.
- [3] J. Y. Campbell and M. Yogo. Efficient tests of stock return predictability. *Journal of Financial Economics*, 81(1):27–60, 2006.
- [4] E. Carlstein. The use of subseries values for estimating the variance of a general statistic from a stationary sequence. *The Annals of Statistics*, 14(3):1171–1179, 1986.
- [5] M. Carrasco and X. Chen. Mixing and moment properties of various GARCH and stochastic volatility models. *Econometric Theory*, 18(1):17–39, 2002.
- [6] Y. Choi, S. Jacewitz, and J. Y. Park. A reexamination of stock return predictability. *Journal of Econometrics*, 192(1):168–189, 2016.
- [7] B. Efron and R. J. Tibshirani. *An introduction to the bootstrap*. Chapman & Hall, 1993.

- [8] J. Fan and Q. Yao. *Nonlinear Time Series: Nonparametric and Parametric Methods*. Springer, New York, 2003.
- [9] B. Fritzenberger. The moving blocks bootstrap and robust inference for linear least squares and quantile regressions. *Journal of Econometrics*, 82(2):235–287, 1997.
- [10] A. Goyal and I. Welch. Predicting the equity premium with dividend ratios. *Management Science*, 49(5):639–654, 2003.
- [11] K. B. Gregory, S. N. Lahiri, and D. J. Nordman. A smooth block bootstrap for quantile regression with time series. *The Annals of Statistics*, 46(3):1138–1166, 2018.
- [12] P. Hall, J. L. Horowitz, and B.-Y. Jing. On blocking rules for the bootstrap with dependent data. *Biometrika*, 82(3):561–574, 1995.
- [13] W. Hendricks and R. Koenker. Hierarchical spline models for conditional quantiles and the demand for electricity. *Journal of the American Statistical Association*, 87(417):58–68, 1992.
- [14] Y. M. Kim and D. J. Nordman. Properties of a block bootstrap under long-range dependence. *Sankhyā Series A*, 73(1):79–109, 2011.
- [15] K. Knight. Limiting distributions for L_1 regression estimators under general conditions. *The Annals of Statistics*, 26(2):755–770, 1998.
- [16] R. Koenker. *Quantile regression*. Cambridge University Press, 2005.
- [17] R. Koenker and G. Bassett Jr. Regression quantiles. *Econometrica*, 46(1).
- [18] R. Koenker and G. Bassett Jr. Robust tests for heteroscedasticity based on regression quantiles. *Econometrica*, 50(1):43–61, 1982.
- [19] R. Koenker and Q. Zhao. Conditional quantile estimation and inference for ARCH models. *Econometric Theory*, 12(5):793–813, 1996.
- [20] A. Kostakis, T. Magdalinos, and M. P. Stamatogiannis. Robust econometric inference for stock return predictability. *The Review of Financial Studies*, 28(5):1506–1553, 2014.
- [21] H. R. Künsch. The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3):1217–1241, 1989.
- [22] S. Lahiri. On second-order properties of the stationary bootstrap method for studentized statistics. In S. Ghosh, editor, *Asymptotics, Nonparametrics, and Time Series*, pages 683–711. Marcel Dekker, New York, 1999.
- [23] S. Lahiri. Consistency of the jackknife-after-bootstrap variance estimator for the bootstrap quantiles of a studentized statistic. *The Annals of Statistics*, 33(5):2475–2506, 2005.

- [24] J. H. Lee. Online supplement to “predictive quantile regression with persistent covariates: IVX-QR approach”. Available at <https://sites.google.com/site/jihyung412/research>. 2014.
- [25] J. H. Lee. Predictive quantile regression with persistent covariates: IVX-QR approach. *Journal of Econometrics*, 192(1):105–118, 2016.
- [26] X. S. Liao, Z. W. Cai, and H. Q. Chen. A perspective on recent methods on testing predictability of asset returns. *Applied Mathematics-A Journal of Chinese Universities*, 33(2):127–144, 2018.
- [27] A. M. Lindner. Continuous time approximations to GARCH and stochastic volatility models. In T. Andersen, R. Davis, J.-P. Kreiss, and T. Mikosch, editors, *Handbook of Financial Time Series*, pages 481–496. Springer, New York, 2009.
- [28] R. Y. Liu and K. Singh. Moving blocks jackknife and bootstrap capture weak dependence. In R. Lepage and L. Billard, editors, *Exploring The Limits of Bootstrap*, pages 225–248. Wiley, New York, 1992.
- [29] T. Magdalinos and P. C. Phillips. Econometric inference in the vicinity of unity. *Singapore Management University, CoFie Working Paper*, 7, 2009.
- [30] A. Maynard, K. Shimotsu, and N. Kuriyama. Inference in predictive quantile regressions. *Unpublished Manuscript*, 2018.
- [31] A. Maynard, K. Shimotsu, and Y. Wang. Inference in predictive quantile regressions. *Unpublished Manuscript*, 2011.
- [32] P. C. Phillips. Towards a unified asymptotic theory for autoregression. *Biometrika*, 74(3):535–547, 1987.
- [33] P. C. Phillips and J. H. Lee. Predictive regression under various degrees of persistence and robust long-horizon regression. *Journal of Econometrics*, 177(2):250–264, 2013.
- [34] P. C. Phillips and J. H. Lee. Robust econometric inference with mixed integrated and mildly explosive regressors. *Journal of Econometrics*, 192(2):433–450, 2016.
- [35] P. C. Phillips and T. Magdalinos. Limit theory for moderate deviations from a unit root. *Journal of Econometrics*, 136(1):115–130, 2007.
- [36] D. N. Politis and J. P. Romano. A circular block-resampling procedure for stationary data. In R. Lepage and L. Billard, editors, *Exploring The Limits of Bootstrap*, pages 263–270. Wiley, New York, 1992.
- [37] D. N. Politis and J. P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

- [38] D. Pollard. Asymptotics for least absolute deviation regression estimators. *Econometric Theory*, 7(2):186–199, 1991.
- [39] S. Portnoy. Asymptotic behavior of regression quantiles in non-stationary, dependent cases. *Journal of Multivariate Analysis*, 38(1):100–113, 1991.
- [40] R. J. Serfling. *Approximation theorems of mathematical statistics*. Wiley, New York, 1980.
- [41] M. M. Siddiqui. Distribution of quantiles in samples from a bivariate population. *J. Res. Nat. Bur. Standards*, 64B:145–150, 1960.
- [42] I. Welch and A. Goyal. A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508, 2008.
- [43] A. Welsh. Kernel estimates of the sparsity function. In Y. Dodge, editor, *Statistical Data Analysis Based on the L_1 -Norm and Related Methods*, pages 369–377. Elsevier, New York, 1987.
- [44] W. B. Wu. Nonlinear system theory: Another look at dependence. *Proceedings of the National Academy of Sciences of the United States of America*, 102(40):14150–14154, 2005.
- [45] Z. Xiao. Quantile cointegrating regression. *Journal of Econometrics*, 150(2):248–260, 2009.
- [46] Z. Xiao and R. Koenker. Conditional quantile estimation for generalized autoregressive conditional heteroscedasticity models. *Journal of the American Statistical Association*, 104(488):1696–1712, 2009.